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Master's Thesis

Master's Degree in Data Science

Regime-Aware Hierarchical Modeling for Intra-day VIX Dynamics

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Abstract

Forecasting short-horizon movements in implied volatility constitutes a persistent challenge in quantitative finance due to non-stationarity, volatility clustering, structural regime changes, and non-linear cross-market dependencies. This study proposes a *novel* regime-sensitive multi-model classification framework for the categorical prediction of intra-day directional movements in the CBOE Volatility Index (VIX). The architecture combines kernel-based discriminative models with non-parametric tree-based learners under a dynamic decision aggregation scheme specifically designed to adapt to heterogeneous volatility regimes.

To operationalize the forecasting framework in a production-grade, real-time environment, this study implements an event-driven cloud-native data infrastructure based on Microsoft Azure Functions. A timer-triggered serverless orchestration layer manages the automated ingestion of multi-asset market data, the computation of derived volatility, term-structure, and cross-market features, and the persistent storage of processed observations within a cloud-hosted MongoDB NoSQL environment. The infrastructure is designed to support low-latency processing of high-dimensional financial time series while ensuring horizontal scalability, fault tolerance, and schema flexibility required by continuously evolving market datasets. Feature generation, validation, transformation, and persistence are fully automated within a modular ETL workflow, enabling reproducible data acquisition and seamless integration with downstream inference pipelines. As a result, the proposed architecture provides an end-to-end deployment framework for continuous volatility forecasting under live market conditions.

To address class imbalance and enhance predictive robustness, targets are defined using quantile-based categorical thresholds on intra-day VIX returns, ensuring balanced class representation across heterogeneous market conditions. Model hyperparameters are optimized within a walk-forward validation framework, preserving temporal dependencies while mitigating overfitting and preventing information leakage during calibration. Performance is subsequently assessed through an out-of-sample backtesting procedure spanning multiple test windows with varying horizons, enabling the evaluation of model stability, robustness, and generalization across structurally distinct market regimes. Empirical results indicate that performance gains are particularly pronounced during high-volatility regimes, highlighting the importance of explicitly incorporating regime sensitivity into volatility forecasting models.Across all evaluated forecasting horizons, the best-performing specification is a logistic regression model, which outperforms more complex non-linear learners, including tree-based ensembles, in out-of-sample backtesting. In addition, it exhibits strong stability across evaluation windows, achieving a median (Q2) weighted F1-score of 0.54 and a substantially lower variability with a standard deviation of 0.081.

Overall, the empirical evidence suggests that the proposed architecture constitutes a statistically robust and computationally scalable framework for short-horizon volatility classification under non-stationary market dynamics. By integrating regime-sensitive machine learning with an event-driven cloud-native data infrastructure, the system achieves consistent out-of-sample generalization across heterogeneous backtesting horizons and structurally distinct volatility regimes. From a systems perspective, the serverless orchestration layer and distributed NoSQL storage architecture provide low-latency feature processing, fault-tolerant execution, and schema adaptability, while enabling seamless horizontal scaling as data dimensionality, asset coverage, and inference frequency increase. These properties establish a reproducible and production-ready computational framework for real-time volatility modeling, systematic execution, and adaptive risk calibration in high-frequency quantitative environments.

Keywords: Regime-Aware Classification; Financial Time-Series Modeling; Quantitative Finance; Walk-Forward Validation; Non-linear Feature Engineering; Cloud-Based Data Pipelines; NoSQL Databases; Concept Drift

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Annotated

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1 Introduction, Objective, and Motivation

1.1 Introduction

Forecasting financial market volatility remains one of the most challenging problems in quantitative finance due to its non-linear, non-stationary, and regime-dependent nature. The VIX [14], introduced by the Chicago Board Options Exchange (CBOE), is widely regarded as a benchmark indicator of implied market volatility and investor sentiment. It is commonly interpreted as a real-time proxy for systemic risk and uncertainty in global financial markets. Intraday fluctuations of the VIX reflect rapid changes in expectations, liquidity conditions, and macroeconomic information flows, making their prediction both complex and economically relevant.

From a computational perspective, the VIX is calculated from a cross-section of S&P 500 index (SPX) options. Specifically, it uses mid-quote prices of a wide range of *out-of-the-money* call and put options with approximately 30 calendar days to maturity. Rather than relying on a single option contract or a particular option-pricing model, the methodology aggregates information across a continuum of strikes using a model-free implied variance framework. This approach extracts the market's consensus expectation of future volatility embedded in option prices.

Formally, the VIX is derived from a discretized version of the risk-neutral variance swap rate. Let K_i denote the strike prices of available SPX options, ΔK_i the interval between adjacent strikes, $Q(K_i)$ the midpoint of the bid–ask quote for each option, F the forward level of the S&P 500 index, K_0 the first strike below the forward price, T the time to expiration expressed in years, and r the risk-free interest rate.

MXEA 2090 Calls Expiring 5/19/2023 with timestamp t = 15:19:30				
Time of Quote	**Bid**	**Ask**	**Spread**	**Mid**
15:19:19.255645	54.8	58.9	4.1	56.85
15:19:19.255967	54.8	59.3	4.5	57.05
15:19:19.822725	54.4	58.9	4.5	56.65
15:19:20.138311	54.6	59.1	4.5	56.85
15:19:20.261043	54.6	59.1	4.5	56.85
15:19:21.588101	54.9	59.1	4.2	57
15:19:21.588945	54.9	59.4	4.5	57.15
15:19:25.951666	54.9	59.4	4.5	57.15
15:19:26.025636	54.9	59.3	4.4	57.1
15:19:26.029053	54.8	59.3	4.5	57.05
15:19:26.444674	54.9	59.3	4.4	57.1
15:19:26.445398	54.9	59.4	4.5	57.15
15:19:27.525609	54.9	59.4	4.5	57.15
15:19:27.527957	49.6	64.6	15	57.1
15:19:28.122755	50.1	65.1	15	57.6
15:19:28.690431	50.1	65.1	15	57.6
15:19:29.907117	50.3	65.1	14.8	57.7
15:19:29.908263	50.3	65.3	15	57.8

Fig. 1: Illustrative bid–ask option quotes used in the computation of the VIX (source: Cboe Global Markets [14]).

The model-free implied variance is computed as

$$\sigma^2 = \frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) - \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2$$

where the summation spans all *out-of-the-money* calls and puts across available strikes. The VIX index is then defined as the square root of the weighted average of the interpolated 30-day forward variances derived from the two nearest option maturities, annualized and expressed in percentage terms:

$$\text{VIX} = 100 \times \sqrt{\sigma^2_{30d}}$$

In practical terms, the VIX can be interpreted as the market's consensus estimate of the annualized standard deviation of S&P 500 returns over the next month, derived directly from option prices. Traditional econometric approaches, such as GARCH-type models, capture certain stylised facts of volatility dynamics but often struggle to adapt to abrupt regime changes and complex non-linear interactions present in financial markets. Machine learning techniques have shown strong potential for modeling financial time series by exploiting high-dimensional data structures and non-linear feature interactions. Nevertheless, short-horizon volatility forecasting remains an open challenge due to continuously evolving macroeconomic conditions, latent regime transitions, and the absence of universally stable predictive relationships across market environments.

1.1.1 Relationship Between Implied and Realised Volatility

Implied volatility indices such as the VIX are commonly interpreted as market-based, forward-looking measures of expected future volatility, as discussed in subsection 1.1, whereas realised volatility captures the ex-post variability of underlying asset returns. Understanding the dynamic interaction between these two volatility measures is therefore of particular relevance, as it provides insight into whether option markets primarily anticipate future volatility conditions or, alternatively, incorporate information contained in recently observed market fluctuations.

To deeply investigate this relationship, Adhikari and Hilliard analyse the dynamic linkage between the VIX and the realised volatility (RV) of the S&P 500 using a multivariate lagged regression framework combined with Granger causality testing [6]. The lag structure of the vector autoregressive specification is selected according to the Bayesian Information Criterion (BIC), ensuring a parsimonious representation of the temporal dependence between both volatility measures.

The empirical results indicate strong evidence that lagged realised volatility contains significant predictive information for subsequent movements in implied volatility. In particular, the first lag of realised volatility exhibits the largest statistical contribution, with a highly significant coefficient and a t-statistic exceeding 9, suggesting that recent realised market fluctuations are rapidly incorporated into option-implied volatility expectations. The joint significance test of lagged RV terms strongly rejects the null hypothesis of no Granger causality, confirming that realised volatility provides incremental explanatory power for the VIX beyond its own autoregressive dynamics.

Variable	Coefficient	Std. Error	t-statistic	p-value
Constant	2.5746	1.7085	1.7085	0.1334
RV_{t-1}	0.7503	0.0786	9.5442	0.0000
RV_{t-2}	0.3359	0.1156	2.9061	0.0041
RV_{t-3}	0.1613	0.1206	1.3381	0.1826
RV_{t-4}	-0.1367	0.1238	-1.1040	0.2709
RV_{t-5}	-0.0239	0.1247	-0.1916	0.8481
RV_{t-6}	0.0536	0.1248	0.4296	0.6680
VIX_{t-1}	0.2019	0.1893	1.0673	0.2874
VIX_{t-2}	0.0573	0.1905	0.3005	0.7640
VIX_{t-3}	-0.3664	0.1896	-1.9338	0.0547
VIX_{t-4}	-0.0319	0.1914	-0.1666	0.8679
VIX_{t-5}	0.1695	0.1837	0.9225	0.3574
VIX_{t-6}	-0.0068	0.1301	-0.0523	0.9581
Granger Test	$F(9, 199) = 1.5206 \quad (p = 0.1426)$			
R^2	0.6386			
Durbin-Watson	1.9990			

TABLE 1: Regression results for testing Granger causality between realised volatility and the VIX (source: Adhikari and Hilliard, International Journal of Financial Markets and Derivatives [6]).

In contrast, when the reverse causality is examined, the authors find no statistically significant evidence that lagged VIX observations improve forecasts of realised volatility once the autoregressive structure of realised volatility is taken into account. The corresponding joint significance test yields an F-statistic of $F(9, 199) = 1.52$ with a p-value of 0.1426, implying that the null hypothesis of no Granger causality cannot be rejected at conventional significance levels.

From an economic perspective, these findings suggest that implied volatility is not entirely exogenous, but instead partially reflects recently observed realised market conditions. This behaviour is consistent with well-documented volatility persistence and clustering effects in financial markets, whereby periods of elevated realised volatility influence subsequent option pricing and volatility expectations.

The mentioned empirical insights provide a direct motivation for the feature engineering strategy adopted in this research, where multiple realised volatility estimators, volatility-of-volatility metrics, and persistence-based market state indicators are incorporated as explanatory variables for short-horizon VIX direction classification.

1.2 Objective

The main objective of this thesis is to develop a regime-aware quantitative framework for the categorical prediction of intraday movements in the CBOE Volatility Index (VIX).

To achieve this objective, a high-dimensional feature space is constructed from historical CBOE VIX data [14], S&P 500 benchmarks, and market microstructure and volatility indicators, ensuring a consistent time-ordered structure suitable for automated processing. A regime-aware component based on latent state modelling is used to detect structural breaks, volatility clustering, and regime transitions, with regime outputs stored in a flexible data layer to enable conditional retrieval and adaptive modelling. At the predictive layer, heterogeneous supervised classifiers are integrated within a generalization framework. Model estimation, hyperparameter optimisation, and evaluation are conducted under a strictly time-dependent validation framework to ensure reproducibility and avoid look-ahead bias, with performance assessed using macro-F1, ROC-AUC, precision, and regime-conditional stability across multiple out-of-sample horizons.

The specific objective is therefore to design and implement a fully automated and extensible predictive system for intraday VIX directionality, operationalised as an end-to-end data-to-decision pipeline within a cloud computing environment.

This is implemented through an event-driven architecture where MongoDB provides flexible non-relational storage and Azure Functions orchestrate automated execution of data ingestion, feature engineering, model training, and inference, ensuring modularity, scalability, and reproducibility. In alignment with the general objective, the framework integrates regime-aware modelling within a unified system designed to maintain predictive consistency under heterogeneous volatility regimes and non-stationary market dynamics, enabling the generation of stable and statistically reliable intraday forecasts.

1.3 Motivation

Modeling intraday volatility dynamics has direct implications for quantitative hedging, derivatives market-making, and real-time risk management. In volatility-sensitive portfolios, short-horizon changes in implied volatility directly affect option valuations, hedge ratios, and higher-order risk sensitivities, including vega, gamma, and vanna exposures. Consequently, the ability to anticipate short-term movements in market-implied volatility can improve hedge rebalancing decisions, reduce transaction costs associated with over-hedging, and mitigate exposure to abrupt volatility repricing.

The CBOE Volatility Index (VIX), as the market-implied measure of expected equity volatility, plays a central role in volatility trading, variance replication, dispersion strategies, and cross-asset risk allocation. Forecasting its intraday directional behavior provides potentially actionable information for volatility arbitrage strategies, dynamic delta-vega hedging, and the management of convexity-sensitive derivatives portfolios operating under rapidly changing market conditions.

From a quantitative modeling perspective, intraday volatility prediction constitutes a challenging learning problem due to the presence of microstructure noise, heteroskedasticity, heavy-tailed innovations, mean-reverting behavior, and latent regime transitions. These characteristics violate the stationarity assumptions underlying many classical econometric models and can significantly degrade the stability of static predictive frameworks when market conditions shift abruptly.

In this context, regime-aware machine learning offers a natural framework for adaptive hedging applications, where model parameters and predictive signals can be conditioned on latent volatility states. By integrating probabilistic regime inference with ensemble-based supervised learning, the proposed framework seeks to identify state-dependent predictive structures and improve robustness across heterogeneous market environments.

The motivation of this research is therefore twofold. From a practical standpoint, it aims to evaluate whether regime-aware predictive models can generate statistically reliable signals that support intraday hedging and volatility risk management. From an academic perspective, it contributes to the growing intersection between quantitative finance and machine learning by investigating how meta-learning architectures can enhance predictive stability, regime adaptability, and risk-sensitive decision making in highly stochastic volatility markets.

2 State of the Art

The modeling and forecasting of financial market volatility constitute a fundamental problem in quantitative finance, given its central role in asset pricing, portfolio allocation, and risk management. In particular, volatility dynamics are a key input for derivative valuation and hedging strategies, as well as for the construction of risk-sensitive investment frameworks employed by institutional market participants and systematic trading desks. Volatility is not directly observable and must therefore be inferred either ex post, through realized measures, or ex ante, through implied metrics derived from option prices. Among the latter, the CBOE Volatility Index (VIX) [14] has emerged as a benchmark proxy for market-implied uncertainty, encapsulating the risk-neutral expectation of future variance over a fixed horizon.

Unlike realized volatility, which is inherently backward-looking, the VIX reflects forward-looking market beliefs and incorporates a variety of components, including volatility risk premia, liquidity effects, and investor sentiment [13, 12]. Consequently, its stochastic dynamics exhibit several stylized features, such as long memory, excess kurtosis, regime persistence, and pronounced non-linearities, particularly at high-frequency and intra-day levels.

The classical econometric framework for modeling volatility is grounded in the Autoregressive Conditional Heteroskedasticity (ARCH) model introduced by Engle [17], and its generalization, the Generalized ARCH (GARCH) model proposed by Bollerslev [11]. These models formalize volatility as a latent conditional variance process driven by past innovations and past variances, successfully capturing volatility clustering and persistence. Numerous extensions have been developed to address empirical regularities observed in financial time series, including asymmetric responses to shocks (e.g., EGARCH [30], GJR-GARCH [20]) and long-memory behavior (e.g., FIGARCH [10]).

Despite their widespread use and strong theoretical grounding, these models rely on parametric specifications that impose restrictive functional forms and distributional assumptions. Moreover, they are primarily designed to model realized volatility and may be ill-suited for capturing the dynamics of implied volatility indices such as the VIX, which embed forward-looking expectations and risk premia [15]. Their limited capacity to incorporate high-dimensional covariates and to model complex non-linear interactions further constrains their applicability in modern data-rich environments, intensified by the complexity of intra-day predictions, where the volatility is intensified by the mentioned non-linear interactions.

To account for structural breaks and regime-dependent behavior, regime-switching models have been extensively studied. The Markov-Switching framework introduced by Hamilton [22] allows model parameters to evolve according to an unobserved discrete state variable governed by a Markov chain. In the context of volatility modeling, such approaches enable the identification of latent regimes typically associated with distinct market conditions, such as tranquil and turbulent periods. Markov-Switching GARCH (MS-GARCH) models combine conditional heteroskedasticity with regime dynamics [21], offering a richer representation of volatility processes. However, these models suffer from significant computational and inferential challenges, including path dependence, likelihood intractability, and parameter proliferation as the number of regimes increases. Furthermore, their flexibility remains constrained by the underlying parametric structure imposed within each regime.

In recent years, the increasing availability of large-scale financial datasets has motivated the adoption of machine learning techniques, which offer a non-parametric alternative for modeling complex and high-dimensional relationships. From a statistical learning perspective [23], these methods aim to approximate unknown functional mappings between predictors and target variables by minimizing empirical risk under minimal structural assumptions.

Linear machine learning models, such as logistic regression, retain a prominent role due to their interpretability and well-understood statistical properties. When combined with regularisation techniques (e.g., Lasso and Ridge penalties), they provide a principled framework for controlling model complexity and mitigating overfitting in high-dimensional settings [36]. Nevertheless, their representational capacity is fundamentally limited to linear or mildly non-linear relationships, even when augmented with feature engineering techniques such as polynomial expansions.

More flexible non-linear models have therefore been proposed to better capture the intricate dependencies present in financial data. Among these, tree-based ensemble methods, particularly gradient boosting algorithms, have achieved state-of-the-art performance in a wide range of tabular data applications [18]. Frameworks such as LightGBM [26] implement efficient gradient boosting schemes that iteratively construct additive models by fitting decision trees to the negative gradient of a specified loss function. These methods are capable of capturing complex, high-order interactions and non-linear effects without requiring explicit feature transformations. Additionally, regularisation techniques such as shrinkage [34, 31], subsampling [31], and tree constraints enhance their generalization capabilities in noisy and heterogeneous environments. Empirical evidence suggests that such models often outperform traditional econometric approaches, including ARIMAX [28] and SARIMAX [27], particularly in classification and forecasting tasks involving large feature sets. However, a key limitation of these approaches lies in their implicit assumption of a stationary and globally homogeneous data-generating process.

A growing body of literature has therefore focused on integrating latent state modeling with machine learning techniques in order to account for regime-dependent dynamics. Hidden Markov Models (HMMs) [33] provide a probabilistic framework for modeling time series with unobserved regime switches, where observed variables are assumed to be generated conditionally on a latent Markov process [32]. In financial applications, HMMs have been widely employed to identify market regimes characterized by distinct statistical properties in returns and volatility [8]. The inferred posterior probabilities of the latent states can be incorporated into predictive models as additional features, effectively enriching the information set and enabling a form of regime-aware learning.

However, incorporating regime probabilities into a single predictive model may not be sufficient to fully capture structural heterogeneity. From a functional approximation standpoint, attempting to model regime-specific relationships within a unified global model can lead to approximation bias, particularly when the underlying mappings differ substantially across regimes. This limitation has motivated the development of regime-dependent or mixture-of-experts approaches, in which separate models are estimated for each regime and combined according to state probabilities [25]. Such frameworks allow for a decomposition of the prediction task into conditionally homogeneous subproblems, thereby improving both interpretability and predictive accuracy.

Within this paradigm, regime-specific modeling enables the adaptation of model complexity to the prevailing market conditions. In low-volatility regimes, characterized by relatively stable dynamics and lower noise levels, simpler and more regularized models may be preferable due to their reduced variance and enhanced interpretability. In contrast, high-volatility regimes often exhibit rapid structural changes, non-linear dependencies, and increased noise, necessitating the use of more flexible and expressive models. This perspective aligns with the broader concept of non-stationary learning and adaptive modeling, where the structure of the predictive system evolves in response to changes in the underlying data-generating process [19].

The present study builds upon these theoretical and methodological developments by proposing a hybrid framework that combines latent regime inference via Hidden Markov Models with both feature-based and model-based adaptation strategies. Specifically, it investigates the transition from regime-aware models, in which inferred state probabilities are incorporated as explanatory variables, to fully regime-specific architectures, where distinct predictive models are trained and deployed conditional on the inferred latent state. By explicitly accounting for regime-dependent dynamics, the proposed approach aims to enhance predictive performance and robustness in the context of intra-day VIX movement forecasting, addressing the challenges posed by non-linearity, noise, and structural heterogeneity in financial markets.

3 Theoretical Background of Techniques and Metrics

The present section introduces the theoretical foundations underlying the proposed modeling framework, with particular emphasis on methodologies tailored to regime-dependent financial time series. The discussion integrates concepts from time series validation, statistical learning, ensemble methods, and latent variable modeling.

3.1 Walk-Forward Validation with Hybrid Expanding-Rolling Recalibration

The estimation, recalibration, and out-of-sample evaluation of the proposed volatility classification architecture are conducted under a recursive **walk-forward validation** framework based on a **hybrid expanding-rolling origin design**. This methodology is specifically designed to address the statistical and structural complexities inherent to financial time series, including temporal dependence, volatility clustering, latent regime persistence, concept drift, distributional non-stationarity, and structural breaks.

Unlike static resampling procedures, the proposed framework performs recursive pseudo-real-time estimation, whereby the predictive architecture is continuously re-estimated as new market observations become available. Consequently, each prediction is generated under the same information constraints that would exist in a live trading environment, ensuring full consistency between model development, backtesting, and real-world deployment.

The adoption of a moving-origin validation framework is motivated by the well-documented limitations of conventional validation methodologies—including random train-test splitting, bootstrap resampling, and k-fold cross-validation—when applied to sequential financial data. These approaches implicitly assume independent and identically distributed (i.i.d.) observations and therefore fail to account for the autocorrelation structure, path dependence, and serial correlation that characterize market dynamics.

Formally, if $\{x_t\}_{t=1}^{T}$ denotes a financial time series, standard random resampling may generate training observations such that:

$$x_{t+h} \in \mathcal{D}_{train}, \qquad x_t \in \mathcal{D}_{test}, \qquad h > 0,$$

thereby violating chronological causality and introducing forward-looking contamination. Such temporal leakage leads to biased parameter estimation and systematically optimistic out-of-sample performance estimates. For this reason, chronological integrity is explicitly preserved throughout the entire validation process.

To address these limitations, the proposed validation architecture combines two complementary estimation regimes:

1. **Initial expanding estimation phase**
2. **Fixed-length rolling recalibration phase**

This hybrid design allows the framework to jointly exploit long-horizon structural information while preserving adaptive sensitivity to short-term distributional changes.

3.1.1 Initial Expanding Training Window

The validation procedure begins with an initial expanding calibration window spanning approximately **thirteen years of historical observations**. This large-sample initialization is intentionally selected to maximize cross-regime statistical representativeness during the early estimation stage.

From a statistical learning perspective, the expanding initialization serves several critical purposes:

- **Estimator stabilization:** large-sample calibration reduces parameter variance and improves numerical conditioning during optimisation.
- **Cross-regime representation:** exposure to multiple macroeconomic cycles, crisis episodes, and volatility compression regimes improves posterior class-boundary estimation.
- **Feature-space calibration:** volatility-derived predictors, cross-asset dependency measures, and regime-sensitive indicators require sufficient empirical support for robust normalization and moment estimation.
- **Latent state identification:** probabilistic regime models, particularly Hidden Markov structures, require adequate state occupancy to obtain statistically meaningful transition estimates.

This initialization phase enables the model to capture low-frequency dependencies, persistent structural relationships, and long-memory effects before entering the adaptive forecasting stage.

3.1.2 Rolling Recalibration Scheme

Following the initialization phase, the framework transitions into a recursive rolling recalibration regime based on a fixed estimation horizon of **500 trading observations**. At each forecasting origin t, the complete predictive architecture is re-estimated using only the most recent information available at time t, after which a one-step-ahead forecast is generated for observation $t+1$. Formally, for each forecasting origin:

$$\mathcal{D}_{train}^{(t)} = \{x_{t-W_r+1}, \ldots, x_t\}, \qquad W_r = 500,$$

$$\mathcal{D}_{test}^{(t)} = \{x_{t+1}\}.$$

The rolling origin is then recursively updated:

$$t \leftarrow t+1.$$

This recursive moving-source evaluation continues until the end of the available sample.

The resulting temporal partitioning is illustrated in Figure 2, where each rolling origin generates an independent pseudo-real-time backtesting segment.

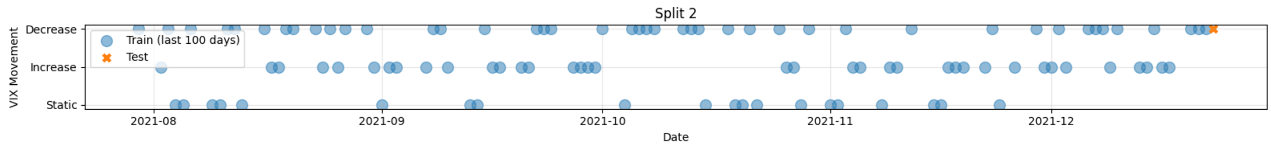

Fig. 2: Sequential 500-observation rolling calibration windows used during recursive walk-forward evaluation. Each split generates an independent one-step-ahead out-of-sample prediction.

The selection of a **500-observation recalibration horizon** is motivated by both statistical efficiency and financial interpretability.

From a statistical perspective, the selected window provides:

From a statistical learning perspective, the selection of a 500-observation rolling calibration horizon provides an effective sample size sufficiently large to ensure stable estimation of high-dimensional parameter spaces, particularly in the presence of non-linear feature transformations, multicollinearity, and ensemble-based interactions. The extended calibration window improves the conditioning of the optimisation problem, facilitates more reliable covariance and correlation structure estimation among highly dependent predictors, and reduces the sensitivity of parameter estimates to local sampling noise. Furthermore, within the regime-detection component, such sample depth ensures adequate state occupancy across latent volatility clusters, allowing transition probabilities, state-dependent moments, and posterior regime assignments to be estimated with lower statistical uncertainty. Overall, the selected horizon contributes to a significant reduction in estimator variance relative to model complexity, thereby improving generalization capacity and out-of-sample parameter stability across successive recalibration cycles.

From a financial perspective, a 500-trading-day horizon approximately spans two complete market years, providing sufficient temporal coverage to capture medium-frequency volatility dynamics that typically evolve beyond short-term market noise. This horizon allows the model to observe complete earnings reporting cycles across multiple sectors, seasonal liquidity effects, and recurrent calendar-driven market anomalies. Additionally, it incorporates the transmission effects associated with monetary policy adjustments, interest-rate normalization cycles, and macroeconomic regime transitions, whose impact on implied volatility often unfolds over several quarters. The selected window also preserves persistent cross-asset dependency structures between equity, fixed-income, commodity, and currency markets, while simultaneously capturing the gradual mean-reversion dynamics that frequently follow systemic volatility shocks. Consequently, the 500-day recalibration horizon offers an empirically robust compromise between structural representativeness and adaptive responsiveness under evolving market conditions.

The empirical relationship between predictive stability and calibration sample size is illustrated in Figure 3, which motivates the selection of sufficiently large rolling estimation windows.

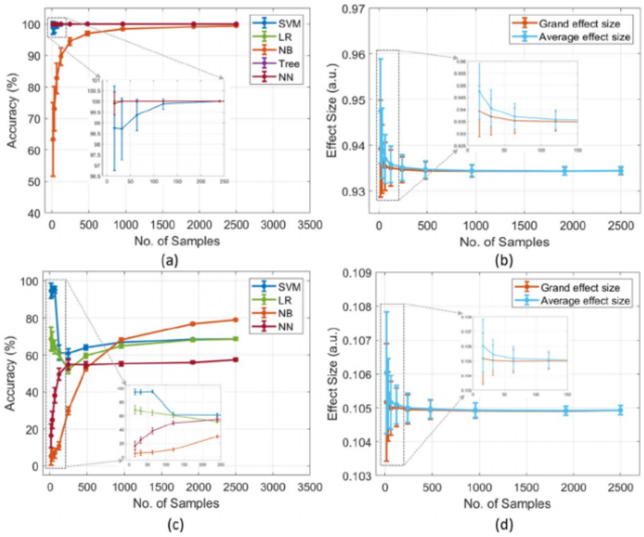

Fig. 3: Sensitivity of predictive stability to training sample size (adapted from Rajput et al.).

A critical advantage of the proposed moving-origin framework is that it generates a sequence of **multiple independent backtesting segments** rather than a single static holdout sample. This property is essential in financial forecasting, where predictive performance is highly sensitive to the specific market regime selected for evaluation.

Let:

$$\mathcal{B} = \{B_1, \dots, B_S\}$$

denote the set of S recursively generated out-of-sample backtesting segments. For any performance metric M, the final reported estimate is computed as:

$$\bar{M} = \frac{1}{S}\sum_{s=1}^{S} M^{(s)}.$$

This repeated evaluation procedure explicitly quantifies sampling variability across low-volatility compression periods, transitional market environments, and extreme stress regimes, thereby mitigating temporal selection bias associated with single-period backtests.

The complete recursive validation process is summarized in Algorithm 1.

Algorithm 1 Walk-Forward Validation with Hybrid Expanding-Rolling Recalibration

Require: Dataset $\mathcal{D}$, Initial Window W_0, Rolling Window W_r, Forecast Horizon H
Ensure: Out-of-Sample Predictions and Performance Metrics

```
t ← W_0
while t + H ≤ |D| do
    (X_train, Y_train) ← D[t − W_r : t]
    (X_test, Y_test) ← D[t + 1]
    Train base learners
    Train meta-learner using posterior probabilities
    Ŷ_{t+1} ← f(X_test)
    Compute metrics:
        ROC-AUC, Macro-F1, Precision, Recall, Confusion Matrix
    Store backtesting results
    t ← t + 1
end while
return aggregated predictions and metrics
```

3.2 Polynomial Logistic Regression

Logistic regression constitutes a discriminative probabilistic classifier belonging to the family of generalized linear models, where conditional class probabilities are parameterized as a log-linear function of the explanatory variables. In the multiclass setting, given an input vector $x \in \mathbb{R}^p$ and a set of discrete classes $k \in \{1, \ldots, K\}$, the posterior probability of class membership is modeled through the softmax transformation:

$$P(y = k \mid x) = \frac{\exp\left(\beta_0^{(k)} + \beta^{(k)\top} x\right)}{\sum_{j=1}^{K} \exp\left(\beta_0^{(j)} + \beta^{(j)\top} x\right)},$$

where $\beta_0^{(k)}$ denotes the class-specific intercept and $\beta^{(k)} \in \mathbb{R}^p$ represents the corresponding parameter vector associated with class k. Under this formulation, the logarithm of the class-odds ratio remains linear in the transformed feature space, yielding a convex optimisation problem with a unique global optimum under standard regularity conditions.

Model parameters are estimated through maximum likelihood, or equivalently by minimizing the regularized multinomial negative log-likelihood:

$$\mathcal{L}(\beta) = -\sum_{t=1}^{N} \log P(y_t \mid x_t) + \lambda \Omega(\beta),$$

where $\Omega(\beta)$ denotes the regularisation functional and $\lambda > 0$ controls the bias-variance trade-off by penalizing excessive parameter complexity. In practice, both L1 (Lasso) and L2 (Ridge) penalization schemes are considered:

$$\Omega_{L1}(\beta) = \|\beta\|_1, \qquad \Omega_{L2}(\beta) = \|\beta\|_2^2.$$

L1 regularisation promotes sparse coefficient structures through implicit variable selection, whereas L2 regularisation stabilizes parameter estimation, mitigates multicollinearity effects, and improves numerical conditioning in highly correlated feature spaces.

Although the canonical logistic formulation is linear with respect to the original predictors, non-linear decision boundaries can be obtained through explicit polynomial feature expansion. Let

$$\phi_d : \mathbb{R}^p \to \mathbb{R}^{p_d}$$

denote a polynomial transformation of degree d, where $d \in \mathbb{N}$ defines the maximum interaction order between explanatory variables. The transformed feature space contains all monomial combinations up to degree d, including pure powers and cross-interaction terms. Consequently, the dimensionality of the expanded space is given by:

$$p_d = \binom{p+d}{d} - 1,$$

which grows combinatorially with both the number of predictors and the selected polynomial degree.

Under this transformation, the multiclass logistic model becomes:

$$P(y = k \mid x) = \frac{\exp\left(\beta_0^{(k)} + \beta^{(k)\top} \phi_d(x)\right)}{\sum_{j=1}^{K} \exp\left(\beta_0^{(j)} + \beta^{(j)\top} \phi_d(x)\right)}.$$

This formulation allows the model to approximate complex non-linear class boundaries while preserving the convexity of the optimisation problem and the probabilistic interpretability of the estimated posterior distributions. Increasing the polynomial degree d enhances representational capacity by enabling the identification of higher-order interactions, curvature effects, and non-linear dependencies that frequently characterize financial datasets. However, larger polynomial degrees also induce a substantial increase in model dimensionality, estimation variance, and overfitting risk, making regularisation an essential component of the learning process.

In the context of volatility regime classification, polynomial logistic regression provides a computationally efficient and statistically interpretable baseline capable of modeling non-linear relationships between macro-financial indicators, market microstructure variables, and latent volatility dynamics. Its probabilistic outputs, convex estimation framework, and controllable model complexity make it particularly suitable for sequential financial forecasting under high-dimensional and non-stationary conditions.

3.3 Gradient Boosting

Gradient boosting is an iterative ensemble learning framework in which a strong predictive model is constructed as an additive expansion of weak learners, typically decision trees. The procedure can be interpreted as functional gradient descent, where each new learner is fitted to the negative gradient of a differentiable loss function.

Formally, the model is defined as:

$$f(x_t) = \sum_{m=1}^{M} \gamma_m h_m(x_t),$$

and updated iteratively as:

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x),$$

where $h_m(x)$ denotes the weak learner at iteration m, γ_m is the step size (learning rate), and M is the total number of boosting rounds.

In this work, LightGBM is employed as the gradient boosting implementation. Let $x_t \in \mathbb{R}^p$ denote the feature vector at time t. LightGBM optimizes the following regularized empirical risk:

$$\mathcal{L} = \sum_t \ell(y_t, f(x_t)) + \Omega(f),$$

where $\ell(\cdot)$ is the multiclass log-loss and $\Omega(f)$ is a structural regularisation term penalising tree complexity (e.g. number of leaves and leaf weights).

At each iteration, the algorithm fits a new tree $h_m(x)$ to the negative gradient of the loss with respect to the current model prediction, thereby performing steepest descent in function space. LightGBM further adopts a leaf-wise tree growth strategy with depth constraints, selecting splits that maximize the reduction in loss.

In this work, CatBoost is employed as an alternative gradient boosting implementation. As in standard boosting frameworks, it constructs an additive model of weak learners (decision trees) trained sequentially to minimise a differentiable loss function under a functional gradient descent scheme.

The model is defined as:

$$f(x_t) = \sum_{m=1}^{M} \gamma_m h_m(x_t),$$

with iterative updates given by:

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x),$$

where $h_m(x)$ is the decision tree fitted at iteration m, γ_m is the learning rate, and M denotes the total number of boosting iterations.

CatBoost optimises the same regularised empirical risk:

$$\mathcal{L} = \sum_t \ell(y_t, f(x_t)) + \Omega(f),$$

where $\ell(\cdot)$ is the multiclass log-loss and $\Omega(f)$ penalises model complexity.

A key distinction of CatBoost lies in its ordered boosting procedure, which reduces prediction shift by computing gradients over permutations that respect temporal ordering, thereby mitigating target leakage and gradient bias in sequential settings. It further incorporates stochastic regularisation via random permutations, feature subsampling, and controlled tree growth, improving robustness under non-stationary and small-sample regimes. Additionally, CatBoost builds oblivious trees, where identical splitting rules are applied at each level, enhancing regularisation and reducing variance in high-noise environments.

3.3.1 Multiclass Probabilistic Output

For multiclass classification, class-wise scores are aggregated as $f_k(x_t)$ for each class $k \in \{0, 1, 2\}$ and mapped to probabilities via a softmax transformation:

$$P(y_t = k \mid x_t) = \frac{\exp(f_k(x_t))}{\sum_{j=0}^{2} \exp(f_j(x_t))}$$

This formulation ensures a properly normalized probabilistic output, enabling consistent evaluation under log-loss and calibration-based metrics.

Regularisation is implicitly enforced through shrinkage (learning rate), row subsampling, column subsampling, and structural constraints on tree growth, improving generalization in high-dimensional, non-stationary financial time series settings.

3.4 Regime-Aware Modeling via Variational Hidden Markov Models

Financial time series are inherently non-stationary and may exhibit structural changes driven by latent state dynamics. Regime-aware modeling frameworks address this property by assuming that the observed data are generated by an unobserved discrete-state stochastic process.

Variational Hidden Markov Models (VHMMs) provide a probabilistic framework for modeling such regime-switching dynamics. Let $\{x_t\}_{t=1}^T$ denote the observed sequence and $\{z_t\}_{t=1}^T$ the corresponding latent state sequence, where $z_t \in \{1, \ldots, K\}$. The latent process evolves according to a first-order Markov chain:

$$P(z_t \mid z_{t-1}) = A_{z_{t-1}, z_t},$$

where A is the transition probability matrix.

Conditional on the latent state, observations are generated from state-dependent emission distributions. In the Gaussian case:

$$x_t \mid z_t = k \sim \mathcal{N}(\mu_k, \Sigma_k).$$

Unlike classical maximum likelihood estimation based on the Expectation-Maximization (EM) algorithm, variational inference approximates the posterior distribution $P(z_{1:T} \mid x_{1:T})$ with a tractable distribution $q(z_{1:T})$ by maximizing the Evidence Lower Bound (ELBO):

$$\mathcal{L}(q, \theta) = \mathbb{E}_q[\log P(x_{1:T}, z_{1:T} \mid \theta)] - \mathbb{E}_q[\log q(z_{1:T})].$$

This variational formulation improves computational tractability and scalability, while providing implicit regularisation through the choice of variational family. It also mitigates issues such as local optima and parameter degeneracy commonly encountered in standard HMM estimation.

From a predictive perspective, the posterior probabilities $P(z_t = k \mid x_{1:t})$ provide a soft assignment of each observation to latent regimes. These probabilities can be used in downstream predictive models, enabling regime-dependent forecasting.

The predictive distribution can be expressed as a mixture over regimes:

$$P(y_t \mid x_t) = \sum_{k=1}^{K} P(y_t \mid x_t, z_t = k) \cdot P(z_t = k \mid x_{1:t}),$$

which represents a probabilistic ensemble of regime-specific models. Alternatively, hard assignments of z_t can be used to train separate models per regime, resulting in a modular modeling structure.

Overall, Variational Hidden Markov Models provide a flexible and theoretically grounded framework for modeling latent regime dynamics in non-stationary time series.

3.5 Ensemble Learning and Stacking

Ensemble learning refers to a broad class of methodologies that combine multiple predictive models with the objective of improving generalization performance relative to individual estimators. From a statistical perspective, the effectiveness of ensemble methods is rooted in the bias–variance decomposition: by aggregating a diverse set of learners, it is possible to reduce estimation variance while preserving or even improving approximation accuracy. This gain is particularly pronounced when the individual models exhibit low correlation and capture complementary structures of the underlying data-generating process.

Among ensemble techniques, stacking (or stacked generalization) constitutes a flexible and theoretically appealing framework for combining heterogeneous predictors. Unlike simpler aggregation methods such as bagging or boosting, which rely on homogeneous base learners and predefined combination rules, stacking adopts a data-driven approach to model combination. Specifically, a set of base learners $\{f_1, \ldots, f_K\}$ is first trained on the original feature space, and their predictions are subsequently used as inputs to a higher-level meta-learner.

Formally, let $f_k(x)$ denote the prediction of the k-th base model. The stacked predictor is defined as:

$$\hat{y} = g\big(f_1(x), f_2(x), \ldots, f_K(x)\big),$$

where $g(\cdot)$ is a meta-model trained to optimally combine the base predictions.

A crucial aspect of stacking lies in the construction of the meta-learning dataset. To prevent overfitting and information leakage, the predictions used to train the meta-learner are typically obtained via out-of-sample procedures, such as cross-validation or, in time series contexts, walk-forward validation. This ensures that the meta-model learns to generalize based on realistic predictive signals rather than in-sample artifacts.

From a functional approximation standpoint, stacking can be interpreted as learning a higher-order mapping over the space of base predictors. This enables the model to capture complex interactions not only between the original features but also among the predictions of different learners. Consequently, stacking is particularly well-suited for settings characterized by non-linear, high-dimensional, and regime-dependent relationships, as commonly observed in financial markets.

In the presence of structural heterogeneity and regime shifts, different base models may exhibit varying levels of performance across market conditions. The meta-learner can exploit this heterogeneity by dynamically weighting the contributions of individual predictors, effectively acting as an adaptive model selection mechanism. This property is especially valuable in regime-dependent frameworks, where no single model is expected to perform optimally across all states of the world.

Overall, stacking provides a principled and highly flexible approach to model aggregation, leveraging both diversity and complementarity among base learners. Its ability to integrate heterogeneous modeling paradigms and to adapt to complex data structures makes it a powerful tool for improving predictive performance in challenging financial forecasting applications.

3.6 Classification Metrics

The predictive performance of the proposed volatility classification framework is evaluated through a multidimensional set of statistical metrics designed to assess not only global predictive accuracy, but also class-level discrimination, probabilistic ranking quality, calibration stability, and robustness across heterogeneous market conditions. In financial time series, model evaluation presents additional challenges compared to standard static classification problems, as predictive performance is inherently sensitive to the specific temporal segment selected for out-of-sample testing. Changes in market microstructure, monetary policy regimes, liquidity conditions, and volatility clustering can substantially alter the statistical properties of the underlying data-generating process, leading to significant variations in model performance across different evaluation periods.

Consequently, a single backtesting sample may produce performance estimates that are highly conditional on the selected market regime and therefore not representative of the model's true generalization capacity. To mitigate this source of temporal selection bias, model performance is evaluated across multiple sequential out-of-sample backtesting windows generated through the walk-forward validation framework described previously. Let $\mathcal{B} = \{B_1, \dots, B_S\}$ denote the set of S independent backtesting segments. For any performance metric M, the final reported estimate is computed as:

$$\bar{M} = \frac{1}{S} \sum_{s=1}^{S} M^{(s)},$$

where $M^{(s)}$ denotes the metric obtained on backtesting segment B_s. This repeated evaluation procedure provides a more statistically robust estimate of predictive quality by explicitly measuring model consistency across low-volatility environments, transitional market phases, and high-volatility stress regimes.

The target variable is formulated as a balanced three-class classification problem, where each volatility regime contains approximately the same number of observations. Let $\mathcal{Y} = \{1, 2, 3\}$ denote the set of volatility classes and let N represent the total number of out-of-sample observations. For each class $k \in \mathcal{Y}$, predictive outcomes are summarized through the confusion matrix, whose entries correspond to true positives (TP_k), false positives (FP_k), true negatives (TN_k), and false negatives (FN_k). This representation provides a complete decomposition of classification outcomes and enables a regime-specific interpretation of prediction errors.

The most direct global performance measure is *classification accuracy*, defined as the proportion of correctly classified observations:

$$\text{Accuracy} = \frac{1}{N} \sum_{k=1}^{K} TP_k.$$

Because the three volatility regimes are intentionally balanced, accuracy constitutes a statistically meaningful first-order measure of predictive effectiveness, as no individual class dominates the empirical distribution. However, despite its interpretability, accuracy alone may conceal systematic regime confusion and does not distinguish between economically asymmetric misclassification errors. For example, incorrectly classifying an extreme-volatility regime as a low-volatility regime may carry substantially greater economic consequences than the inverse error.

To obtain a more granular assessment of predictive quality, class-specific *precision* and *recall* are computed for each volatility regime k:

$$\text{Precision}_k = \frac{TP_k}{TP_k + FP_k}, \qquad \text{Recall}_k = \frac{TP_k}{TP_k + FN_k}.$$

Precision measures the reliability of positive regime assignments, quantifying the proportion of predicted observations belonging to regime k that are effectively correct. In the context of volatility forecasting, high precision reduces the generation of false regime signals, thereby limiting unnecessary portfolio rebalancing, excessive transaction costs, and spurious hedging adjustments.

Conversely, recall measures the sensitivity of the classifier, quantifying its ability to identify all observations truly belonging to a given volatility regime. High recall is particularly relevant for stress regimes, where missed detections may lead to inadequate risk exposure, under-hedging, and delayed portfolio protection.

Since precision and recall often exhibit an inverse relationship, both metrics are jointly summarized through the *F1-score*, defined as their harmonic mean:

$$F1_k = 2 \cdot \frac{\text{Precision}_k \cdot \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}.$$

The harmonic formulation penalizes extreme imbalances between both components and provides a robust measure of class-level predictive quality, particularly when regime transitions generate asymmetric economic risks.

To aggregate class-level performance across the three volatility regimes, macro-averaging is employed. For any class-dependent metric M_k, the macro-average is defined as:

$$M_{\text{macro}} = \frac{1}{K}\sum_{k=1}^{K} M_k, \qquad K = 3.$$

Because all classes are approximately equally represented, macro-averaging ensures that each volatility regime contributes symmetrically to model selection, preventing stable market conditions from disproportionately influencing performance estimates.

Beyond discrete classification metrics, the probabilistic discrimination capacity of each model is evaluated through the *Receiver Operating Characteristic* (ROC) framework. For each volatility regime k, a one-versus-rest decomposition transforms the multiclass problem into three independent binary discrimination tasks. For a given probability threshold $\tau \in [0, 1]$, the classifier assigns regime k whenever:

$$P(y = k \mid x) \geq \tau.$$

By continuously varying τ, the ROC curve is obtained by plotting the *True Positive Rate* (TPR) against the *False Positive Rate* (FPR):

$$\text{TPR}_k = \frac{TP_k}{TP_k + FN_k}, \qquad \text{FPR}_k = \frac{FP_k}{FP_k + TN_k}.$$

Geometrically, the ROC curve characterizes the complete trade-off between sensitivity and false alarm generation across all possible decision thresholds. A curve concentrated near the upper-left corner indicates strong class separability, whereas a diagonal trajectory corresponds to random classification.

The associated *Area Under the Curve* (AUC) is defined as:

$$\text{AUC}_k = \int_0^1 ROC_k(u)\, du,$$

where values close to 1 indicate near-perfect probabilistic discrimination, while values around 0.5 correspond to random predictive behavior.

To summarize multiclass probabilistic performance, the macro-averaged ROC-AUC across the three one-versus-rest curves is computed:

$$\text{AUC}_{\text{macro}} = \frac{1}{K}\sum_{k=1}^{K} \text{AUC}_k.$$

Unlike threshold-dependent metrics, ROC-AUC evaluates the ranking quality of the predicted posterior probabilities independently of any specific classification cutoff. This property is particularly relevant in financial applications, where posterior probabilities are often incorporated directly into dynamic allocation rules, volatility overlays, position sizing mechanisms, and adaptive hedging systems.

Taken together, the confusion matrix, repeated multi-period backtesting evaluation, macro-averaged precision, recall, F1-score, and ROC-AUC provide a statistically rigorous and economically meaningful evaluation framework capable of assessing not only predictive accuracy, but also regime separability, temporal stability, probability calibration, and robustness under evolving market conditions.

4 Dataset and Feature Engineering

The dataset used in this study consists of approximately twenty years of historical financial market data spanning multiple asset classes and volatility indicators, sufficient to have a direct impact on the ability to explain market volatility. The primary variables are derived from the CBOE Volatility Index (**VIX**), the S&P 500 Index (**SPX**), and the ICE BofA MOVE Index (**MOVE**), which respectively capture implied equity volatility, underlying equity market dynamics, and bond market volatility.

In addition to the mentioned primary volatility benchmarks, the dataset is incorporating cross-asset financial indicators including the U.S. Dollar Index (**DXY**), gold prices (**GOLD**), crude oil prices (**OIL**), and corporate bond ETFs such as the iShares High Yield Corporate Bond ETF (**HYG**) and the iShares Investment Grade Corporate Bond ETF (**LQD**). Mentioned variables provide additional macro-financial context, enabling the proposed model to capture systemic risk signals and cross-market transmission effects between equities, fixed income, commodities, and foreign exchange markets.

The data pipeline is implemented through a cloud-based automated infrastructure built using Azure Functions (subsection 4.3). A timer-triggered serverless function retrieves market data from financial APIs, performs feature computation, and stores the processed observations in a cloud-hosted MongoDB No-SQL database.

Data preprocessing of the dataset involved the following steps:

- **Column cleaning:** Removal of irrelevant columns such as dividends, stock splits, and non-informative volume fields for volatility indices.
- **Data alignment and merging:** Market data from SPX, VIX, MOVE, and cross-asset indicators were merged using the date index to construct a unified time-series dataset.
- **Return calculations:** Logarithmic and simple returns were computed for SPX and VIX in order to capture short-horizon market dynamics and volatility clustering effects.
- **Realised volatility estimation:** Rolling standard deviations of SPX returns were calculated over 5, 10, and 21-day windows and annualized to represent short-term and monthly realised volatility regimes.
- **Volatility-of-volatility features:** Rolling standard deviations of VIX returns over 5-day, 10-day, and 21-day windows were computed to capture fluctuations in implied volatility dynamics.
- **VIX technical indicators:** Additional technical indicators were constructed, including moving averages (5-day, 10-day, 20-day), rolling standard deviations, lagged VIX values, and percentile ranks to capture momentum and mean-reversion behaviour in volatility markets.
- **Gaps and overnight effects:** Opening price gaps for SPX and VIX were computed as the difference between the current opening price and the previous closing price, capturing overnight information shocks.
- **Drawdown metrics:** Maximum cumulative drawdowns of the S&P 500 were calculated in order to capture downside market stress and systemic risk episodes.
- **Momentum indicators:** Multi-horizon momentum signals (1-month, 3-month, and 6-month) based on SPX returns were included to capture medium-term market trends.
- **Mean-reversion signals:** Deviations of VIX from its short-term moving averages were computed to capture potential volatility mean-reversion effects.
- **Implied vs realised volatility measures:** Ratios and spreads between VIX levels and realised volatility estimates were constructed to capture the volatility risk premium embedded in option prices.
- **Term-structure indicators:** The slope of the VIX futures term structure was approximated using the spread between VIX and VIX3M indices as well as contango measures, which capture changes in volatility expectations across maturities.
- **Cross-market indicators:** Ratios between equity and bond volatility measures (VIX/MOVE), as well as rolling correlations between SPX and VIX returns, were computed to capture systemic stress signals.
- **Non-linear transformations:** Squared transformations of key volatility variables were introduced to allow the model to capture potential non-linear relationships.

The final dataset(subsection 4.1) also contains a set of engineered features designed to capture several key stylised facts of financial markets, including volatility clustering, volatility-of-volatility dynamics, cross-asset contagion, and regime-dependent behaviour in order to increase the explain ability before the dimensionality reduction process. These features form the input representation used by the posterior model development stage.

4.1 Feature Description

Variable	Description
Open_SP500	Opening price of the S&P 500 Index (decision-time observable).
Open_VIX	Opening price of the CBOE Volatility Index.
Open_MOVE	Opening price of the MOVE Index (bond volatility).
Open_DXY	Opening price of the US Dollar Index (proxy for pre-market FX conditions).
Open_GOLD	Opening price of gold futures (global risk sentiment proxy).
Open_OIL	Opening price of crude oil futures.
Open_HYG	Opening price of iShares High Yield Corporate Bond ETF.
Open_LQD	Opening price of iShares Investment Grade Corporate Bond ETF.
DXY_overnight	Overnight return of DXY from previous close to current open.
GOLD_overnight	Overnight return of gold prices.
OIL_overnight	Overnight return of oil prices.
Drawdown	Cumulative SPX drawdown relative to its historical max.(using lagged prices).
Momentum_1M, 3M, 6M	SPX returns over 1, 3, and 6 months using only past information.
RV_5d, RV_10d, RV_21d	Realized volatility of SPX returns computed from lagged returns (no forward-looking).
RV_21d_Sq	Squared 21-day SPX realized volatility (captures convex effects).
VIX_Vol_5d, VIX_Vol_10d, VIX_Vol_21d	Rolling standard deviation of VIX returns (volatility-of-volatility).
VIX_Lag1, Lag2, Lag5	Lagged VIX levels capturing persistence and autocorrelation.
VIX_MA_5, MA_10, MA_20	Moving averages of VIX computed using lagged values (trend signals without leakage).
VIX_STD_5, STD_10	Rolling standard deviations of lagged VIX (volatility regime indicators).
VIX_Percentile	Rolling 1-year percentile rank of lagged VIX (relative positioning).
VIX_Zscore	Standardized VIX relative to rolling mean and standard deviation (lagged inputs).
VIX_Zscore_Sq	Squared Z-score capturing non-linear effects.
VIX_MeanRev	Deviation of VIX from its lagged moving average (mean-reversion signal).
IV_RV_Ratio	Ratio of lagged VIX to SPX realized volatility (implied vs realized risk).
VIX_RV_Spread	Spread between lagged VIX and realized volatility.
VIX_Trend	Difference between short- and long-term exponential moving averages of lagged VIX.
VIX3M_Spread	Spread between lagged 3-month VIX and spot VIX (term structure).
VIX_Contango	Term-structure contango measure using lagged values.
VIX_MOVE_Ratio	Ratio of lagged VIX to lagged MOVE index (cross-asset volatility signal).
SPX_VIX_Corr_21d	Rolling correlation between lagged SPX and VIX returns.
SPX_Volume_Norm	Normalized SPX trading volume relative to its historical average.
SPX_Gap, VIX_Gap	Overnight gap between previous close and current open.
Intraday_VIX_Move	Target categorical variable: 0 = neutral, 1 = upward move, 2 = downward move.

TABLE 2: Feature set constructed using information available at or before market open, ensuring no look-ahead bias producing data-leakage in the prediction of intra-day VIX movements.

4.2 Dependent Variable

The target variable, `Intraday_VIX_Move`, is defined as a three-class categorical variable based on empirical tertile thresholds of the intraday return distribution of the VIX. The proposed classification framework is designed to identify directional volatility regimes while maintaining a balanced class distribution across the prediction task.

- **0 – Static:** Intraday returns within the central tercile (33^{rd}–66^{th} percentile, $-3.36\% <$ VIX$_{inc}$ $< 0.76\%$), representing relatively stable volatility conditions.
- **1 – Increase:** Intraday returns above the upper tercile ($> 66^{th}$ percentile, VIX$_{inc}$ $> 0.76\%$), corresponding to significant upward volatility movements.
- **2 – Decrease:** Intraday returns below the lower tercile ($< 33^{rd}$ percentile, VIX$_{inc}$ $< -3.36\%$), representing significant downward volatility movements.

4.3 Cloud Storage and Data Pipeline

To support reproducible model development, low-latency inference, and scalable deployment under real-time market conditions, the data infrastructure is implemented using a cloud-native architecture composed of MongoDB Atlas for persistent storage and Azure Functions for serverless orchestration. This design enables asynchronous data processing, schema flexibility, and distributed execution of the end-to-end quantitative pipeline.

The historical dataset is stored in a document-oriented NoSQL database, where each document represents a single trading session and contains the complete set of engineered features, target labels, regime states, and model metadata. This document-based representation preserves temporal consistency while allowing efficient retrieval of sequential observations required for rolling-window training, regime inference, and intraday forecasting.

Data ingestion and preprocessing are orchestrated through an event-driven ETL pipeline deployed as stateless Azure Functions. The pipeline is automatically triggered shortly after market opening (09:30 EST), enabling near real-time model updates and predictive inference under live market conditions. The execution workflow consists of the following stages:

1. **Incremental Market Data Acquisition:** Market observations from SPX, VIX, MOVE, and auxiliary macro-financial indicators are retrieved using a delta-based synchronization mechanism, processing exclusively newly available records since the previous execution cycle. This minimizes I/O overhead and reduces redundant computation.
2. **Feature Extraction and Transformation:** The raw observations are transformed into a high-dimensional feature space including realized volatility estimators, momentum and mean-reversion indicators, overnight gap measures, cross-asset spread dynamics, standardized volatility scores, and non-linear interaction terms, as specified in Section 2.
3. **Persistent Storage and Versioning:** The processed feature vectors are serialized into JSON documents and written to the MongoDB cluster using idempotent upsert operations, ensuring deterministic state synchronization, fault tolerance, and historical version consistency.
4. **Rolling Model Update:** The predictive framework is re-estimated using all observations available up to the previous market close, allowing the model parameters to adapt to evolving volatility regimes and structural changes in market microstructure.
5. **Regime Inference and Predictive Classification:** The updated pipeline performs posterior regime decoding through the Variational Gaussian Hidden Markov Model, followed by supervised ensemble inference to generate probabilistic forecasts for the *static*, *increase*, and *decrease* volatility states.
6. **Signal Dissemination:** Forecast outputs, regime probabilities, and performance diagnostics are distributed through an automated messaging interface, enabling low-latency delivery of actionable signals for downstream trading and hedging applications.

From a systems perspective, the proposed architecture follows a fully decoupled and horizontally scalable design, where storage, feature engineering, model estimation, and inference layers operate independently through event-driven communication. This modularity enables isolated component updates, independent resource allocation, and seamless integration of additional predictive models without disrupting operational continuity.

The combination of serverless compute resources and schema-flexible NoSQL storage provides an infrastructure capable of supporting high-frequency quantitative workflows under production constraints. In the context of volatility forecasting and intraday hedging applications, this architecture ensures deterministic data lineage, low operational latency, and adaptive scalability as both feature dimensionality and market data throughput increase over time.

5 Exploratory Data Analysis (EDA)

5.1 Descriptive Statistics

The dataset comprises daily observations of the CBOE VIX index along with a broad set of multi-asset financial indicators (subsection 4.1), including equity, fixed income, commodity, and currency variables. Features based on these indicators have also been designed, naturally providing time-series explanatory power. The primary objective is to model intra-day VIX movements, transforming them into a categorical target variable based on percentage changes defined by thresholds according to the tertiles distribution (subsection 4.2).

- **Positive skewness:** The distribution of VIX returns exhibits pronounced asymmetry, characterized by more frequent and larger upward spikes relative to downward movements. This behavior reflects the inherently asymmetric nature of market stress, where volatility tends to increase abruptly during periods of uncertainty.
- **Excess kurtosis:** The return distribution displays significant leptokurtosis, indicating heavy tails and an elevated likelihood of extreme volatility realizations. This empirical property is consistent with crisis-driven market dynamics and motivates the adoption of a regime-based modeling framework.
- **Volatility clustering:** Volatility exhibits strong temporal dependence, with high-volatility episodes tending to cluster and persist over time, followed by extended periods of relatively low volatility. This stylized fact is indicative of conditional heteroskedasticity in financial time series.

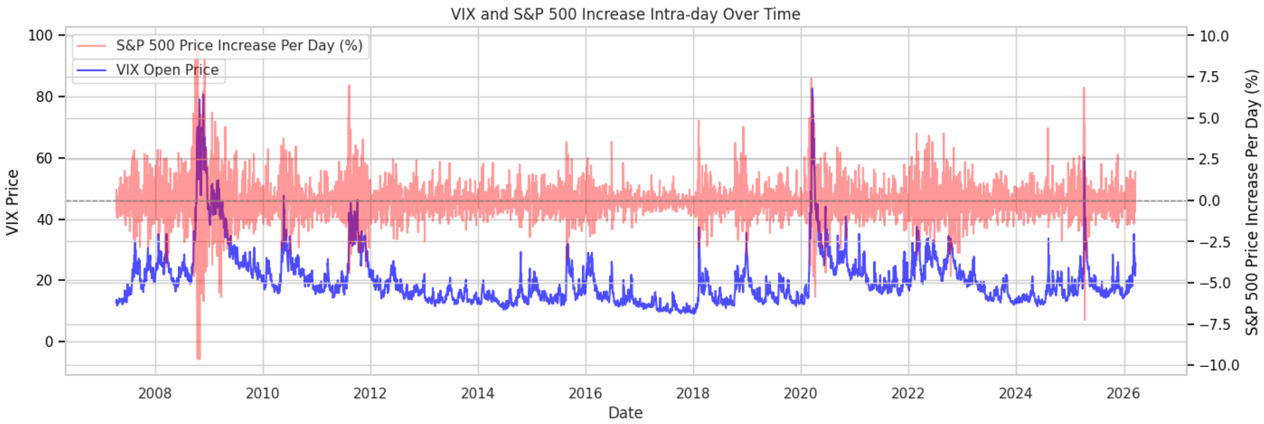

Fig. 4: Intra-day co-movement between VIX and S&P 500 returns over time

While the unconditional mean of VIX returns remains close to zero, consistent with mean-reverting dynamics, higher-order moments vary significantly across subsamples, showing presence of structural regime shifts.

5.2 Distributional Analysis

The distribution reveals certain characteristics:

- A strong concentration of observations around zero, indicating that most intra-day movements are relatively small.

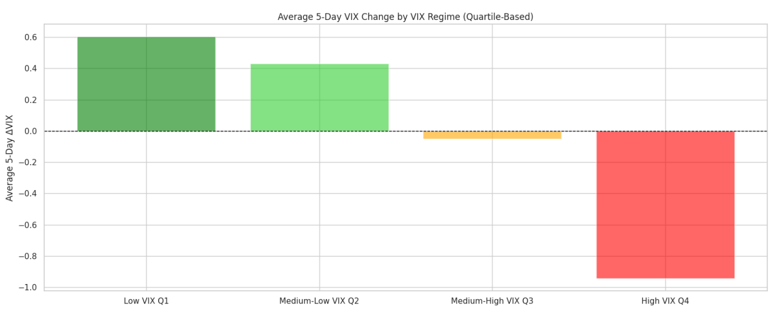

Fig. 5: VIX Quartile Distribution

- Pronounced right tails, capturing abrupt volatility spikes during market stress episodes, switching rapidly from different regimes of the financial season, expressing the need of regime-based models.
- Significant deviations from normality, confirming that Gaussian assumptions are inadequate for modeling VIX dynamics.

5.3 Seasonal Dynamics Decomposition of VIX

This section presents the time-series decomposition of the VIX into trend, seasonal, and residual components.The seasonal component reveals a clear and persistent cyclical structure, despite the widely held assumption that volatility indices are largely driven by unpredictable shocks. This finding partially confirms that part of the VIX dynamics is systematically driven by recurring market mechanisms.

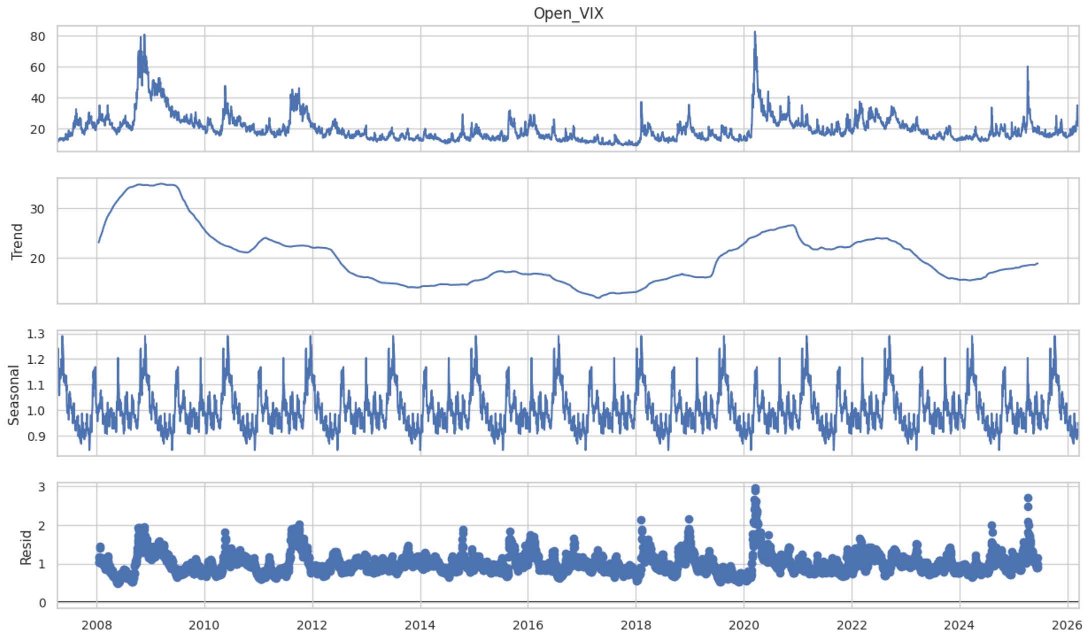

Fig. 6: VIX Seasonal Decomposition

- **High-frequency cyclicality**: The seasonal component exhibits regular oscillations at short horizons, indicating the presence of systematic intra-week or intra-month effects. These cycles are stable across the sample period, suggesting structural repetition rather than noise, highly affected by the seasonal stage of economic trend.
- **Amplitude stability**: Unlike the trend component, which varies significantly across macroeconomic regimes, the seasonal fluctuations remain relatively bounded. This indicates that seasonality operates as a second-order effect superimposed on broader volatility regimes.
- **Mean-reverting oscillations**: The seasonal component exhibits recurrent fluctuations around a stable long-run mean, reinforcing the interpretation of implied volatility as a mean-reverting process, consistent with the empirical findings of Ielpo and Simon, who document statistically significant mean-reversion properties in index-implied volatility surfaces [24]. In addition, the observed asymmetry in the seasonal pattern suggests the presence of persistent option positioning effects, where downside volatility expansions appear bounded by concentrated *put*-side hedging demand (*put wall*), while volatility compression tends to stabilize around *call*-dominated positioning zones (*call floor*). This interpretation is consistent with evidence that call and put option order imbalances contain distinct informational content and influence subsequent volatility formation through dealer inventory rebalancing and gamma-hedging flows [16, 29].

From a modeling perspective, the presence of a stable seasonal component has several implications on the development of models:

- Models that ignore seasonality may systematically misestimate short-term volatility dynamics, depending on the origins of the repeatability of the seasonality.
- Seasonal adjustments or explicitly engineered cyclical features may improve predictive performance, already taken in consideration on the feature engineering section (subsection 4.1).
- The interaction between seasonality and volatility regimes may introduce non-linear effects that are not captured by linear models, it will be explored via distance correlation calculations (subsubsection 5.4.2).

Lastly, the residual component shows that, after removing trend and seasonal effects, the remaining variation is dominated by irregular spikes. These residuals correspond to exogenous shocks and highlight the inherently unpredictable component of volatility dynamics and regime shifts, already introduced in the subsubsection 1.1.1.

5.3.1 Autocorrelation and Persistence in VIX Dynamics

A key stylized fact of volatility indices is their strong temporal dependence. As shown by the autocorrelation function (ACF) of VIX levels and returns, there exists a pronounced persistence structure across short and medium horizons, already evident in the seasonal decomposition.

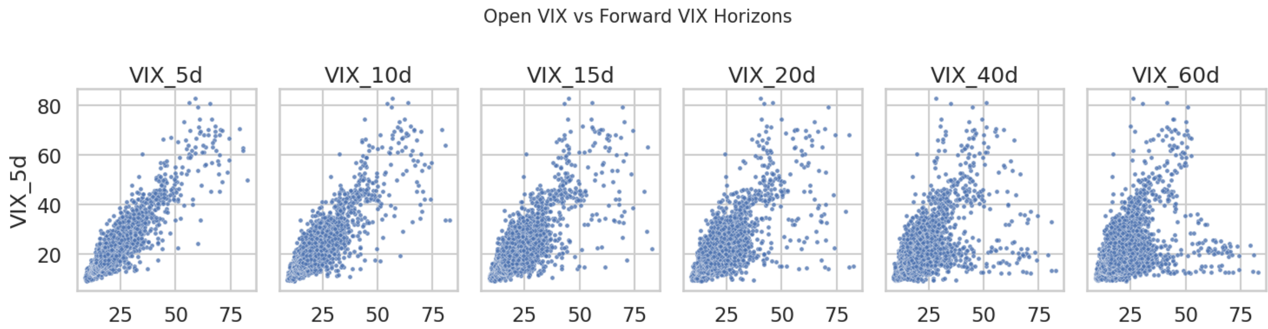

Fig. 7: Autocorrelation Between VIX Shifts

From a probabilistic standpoint, the persistence is consistent with the well-documented clustering of volatility and the slow decay of autocorrelations in second moments, typically associated with long-memory or fractionally integrated processes. Importantly, within this framework, extreme events, often conceptualized as *black swans*[35] do not constitute a dominant component of the realized (ex post) volatility dynamics. Instead, they manifest as rare, high-impact deviations that lie in the tails of the return distribution and are not systematically captured by the central tendency or the bulk behaviour of non-realized volatility measures.

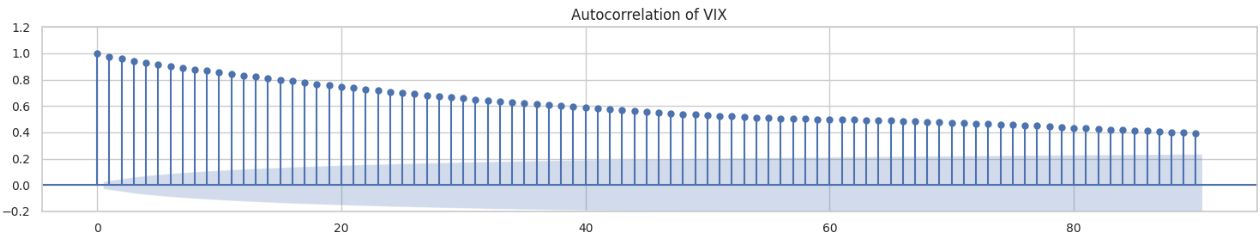

Fig. 8: ACF

More formally, the empirical distribution of realized volatility exhibits excess kurtosis and conditional heteroscedasticity, yet the contribution of tail events to the overall variance decomposition remains limited in frequency. Mentioned insight implies that while such events significantly affect higher-order moments and risk metrics (e.g., Value-at-Risk or Expected Shortfall), they do not materially alter the observed persistence structure in the ACF of volatility indices.

Consequently, the temporal dependence observed in implied volatility indices such as the VIX is primarily driven by endogenous market dynamics (contemplated in subsection 4.1), such as volatility clustering, leverage effects, and mean-reverting stochastic variance processes rather than by sporadic, exogenous shocks of extreme magnitude.

In this sense, black swan events should be interpreted as discontinuities superimposed on an otherwise persistent and mean-reverting volatility process, rather than as intrinsic drivers of the autocorrelation structure of non-realized volatility. This interpretation is consistent with the residual component of the time-series decomposition (subsection 5.3), which primarily captures transitory, idiosyncratic shocks and does not display significant serial dependence, thereby reinforcing the view that persistence is driven by the systematic components of the process rather than by extreme, low-frequency events.

5.4 Correlation Structure: Linear and Nonlinear Dependencies

To characterize the dependence structure among the explanatory variables, are computed both the Pearson correlation matrix (subsubsection 5.4.1) and the distance correlation (dCor) matrix (subsubsection 5.4.2). While Pearson correlation captures linear dependence, distance correlation provides a more general measure capable of detecting arbitrary non-linear associations, complementing the insights presented by Pearson Correlation.

In order to enhance interpretability and focus on economically meaningful relationships, both matrices are threshold to display only absolute values exceeding 0.3.

5.4.1 Pearson Correlation

The following figure reports the Pearson correlation matrix. The extracted insights indicate a pronounced block structure, particularly among volatility-related features. Variables derived from the VIX, including lagged values, moving averages, rolling standard deviations, and volatility-of-volatility measures, exhibit strong pairwise correlations. This reflects the fact that many of these transformations are constructed from overlapping information sets, leading to substantial multi-collinearity. This figure suggest that a principal component analysis can reduce the dimensionality of the dataset in a high degree without a significant lost of the explain-ability of the dependent feature, due to the high correlation between groups of dependent features.

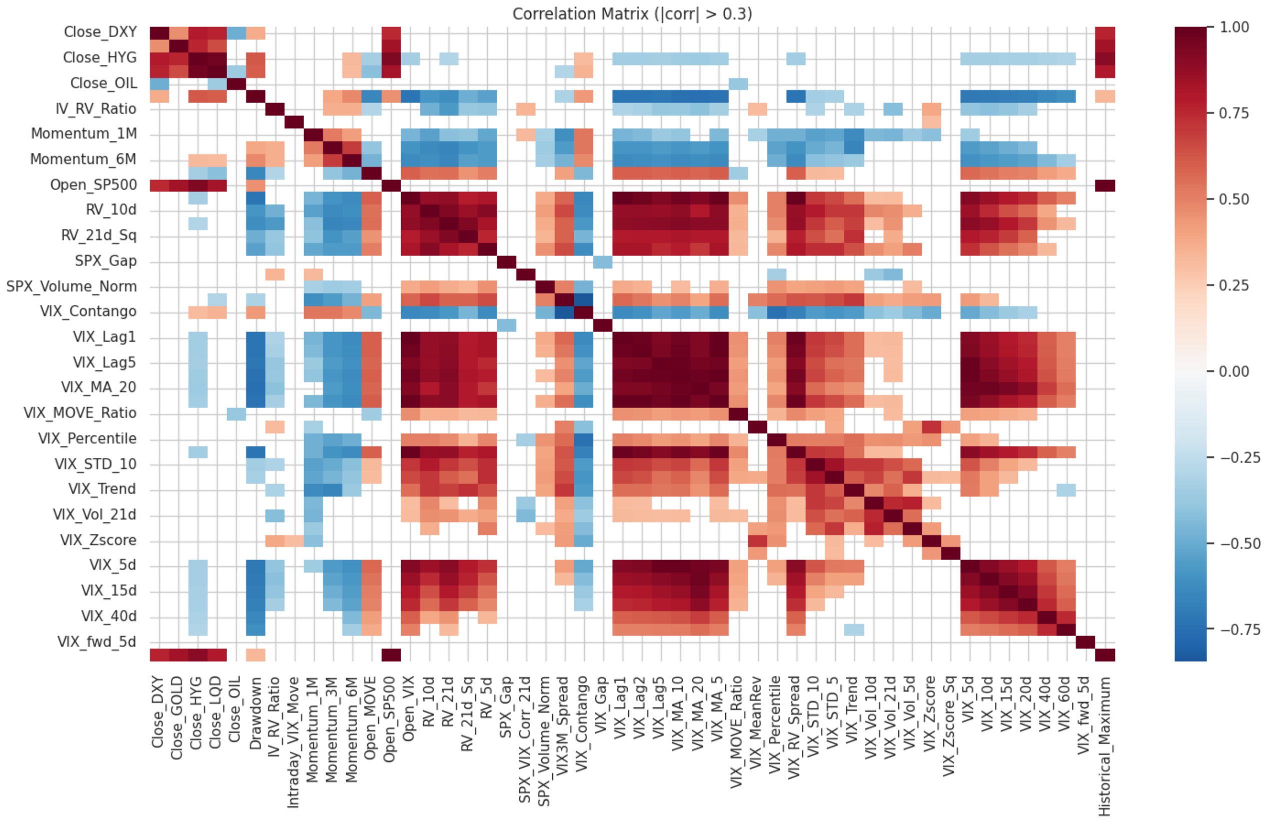

Fig. 9: Pearson Correlation Matrix (|corr| > 0.3 Mask Applied)

In particular, the correlation between contemporaneous VIX-derived features and their lagged counterparts confirms the presence of a strong autoregressive component. This persistence is consistent with the slow decay of volatility shocks and reinforces the relevance of including lagged variables in the feature space. Similarly, realized volatility measures (e.g., 10-day and 21-day windows) form a tightly connected cluster, indicating that different horizon choices capture closely related dynamics as seen in the previous subsubsection 5.3.1.

Cross-asset variables, including equity indices, credit spreads, and macro-financial indicators, display comparatively weaker linear correlations with VIX-based features. However, their relationships are not negligible and tend to strengthen during periods of elevated volatility, confirming that linear correlation may understate their true dependency structure.

5.4.2 Distance Correlation (DCOR)

To address the previous seen limitation, the next correlation presents the distance correlation matrix. Unlike Pearson correlation, distance correlation is defined such that it equals zero if and only if the variables are statistically independent. As a result, it captures both linear and non-linear forms of dependence, complementing the possible unseen insights firstly presented by the previous section subsubsection 5.4.1.

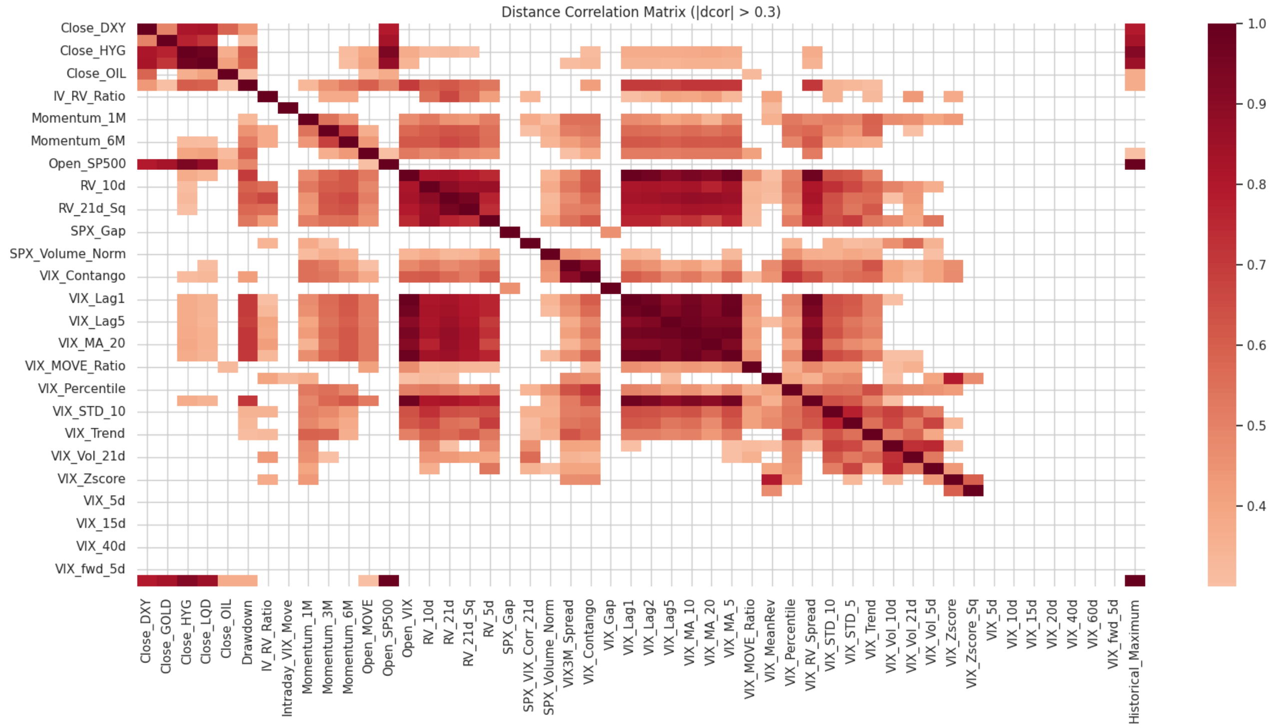

Fig. 10: DCOR Matrix (|dcorr| > 0.3 Mask Applied)

The dCor matrix reveals a substantially richer dependency structure. Several pairs of variables that exhibit weak linear correlation display significant distance correlation, indicating the presence of non-linear interactions. This is particularly evident in features related to volatility spreads, ratios, and standardized transformations, which tend to respond asymmetrically to market conditions. Furthermore, cross-asset linkages become more pronounced under distance correlation, suggesting that macro-financial variables influence VIX dynamics through non-linear transmission channels, especially during stress regimes.

From a statistical perspective, the coexistence of strong linear dependencies within feature groups and non-linear dependencies across groups has important implications. The high degree of multicollinearity among volatility-derived features implies that linear models may suffer from unstable coefficient estimates and reduced interpretability. At the same time, the presence of non-linear dependence structures indicates that purely linear specifications are likely to omit relevant interactions.

Consequently, these findings provide support for the adoption of flexible modeling frameworks capable of handling both redundancy and non-linearity. In particular, tree-based ensemble methods and other non-parametric approaches are well-suited to exploit such structures without requiring explicit functional form assumptions.

Overall, the correlation analysis highlights that the feature space is characterized by a combination of strong intra-group linear dependence and significant cross-group non-linear interactions. This dual structure reinforces the need for modeling approaches that can jointly account for persistence, multicollinearity, and non-linear effects in the prediction of intra-day VIX movements.

Overall, the decomposition (subsection 5.3) complemented by the correlation matrixes confirms that VIX dynamics can be understood as the combination of three distinct forces: a slow-moving trend driven by macroeconomic conditions, a stable seasonal structure reflecting recurring market behaviour, and a residual component capturing unexpected shocks.

5.5 Dimensionality Reduction via Principal Component Analysis (PCA)

To address the high degree of multicollinearity observed among the explanatory variables and to reduce the dimensionality of the feature space, Principal Component Analysis (PCA) is applied to the standardized dataset.

Let $\mathbf{X} \in \mathbb{R}^{n\times p}$ denote the matrix of standardized features, with $p = 41$. PCA constructs a set of orthogonal linear combinations of the original variables:

$$\mathbf{Z} = \mathbf{X}\mathbf{W}$$

where $\mathbf{W}$ contains the eigenvectors of the covariance matrix of $\mathbf{X}$. The resulting principal components maximize the explained variance under orthogonality constraints.

In the study, the dimensionality of the dataset is reduced from 41 to 14 principal components, corresponding to approximately 34.15% of the original feature (subsection 4.2). The selection of the number of components is based on the cumulative explained variance criterion.

The explained variance ratio of the first components is given by:

$$(0.36\ 0.12\ 0.08\ 0.05\ 0.04\ 0.03\ 0.03\ 0.03\ 0.02\ 0.02\ 0.02\ 0.02\ 0.02\ 0.02)$$

which results in a cumulative explained variance of 87%.

The presented results indicate that a relatively small number of orthogonal factors captures the majority of the variability in the dataset. In particular, the first principal component alone explains 36% of the total variance, suggesting the presence of a dominant common factor, likely associated with overall market volatility conditions.

As presented in the following Figure, the loading matrix of the principal components. The loadings provide insight into the contribution of each original variable to the extracted components.

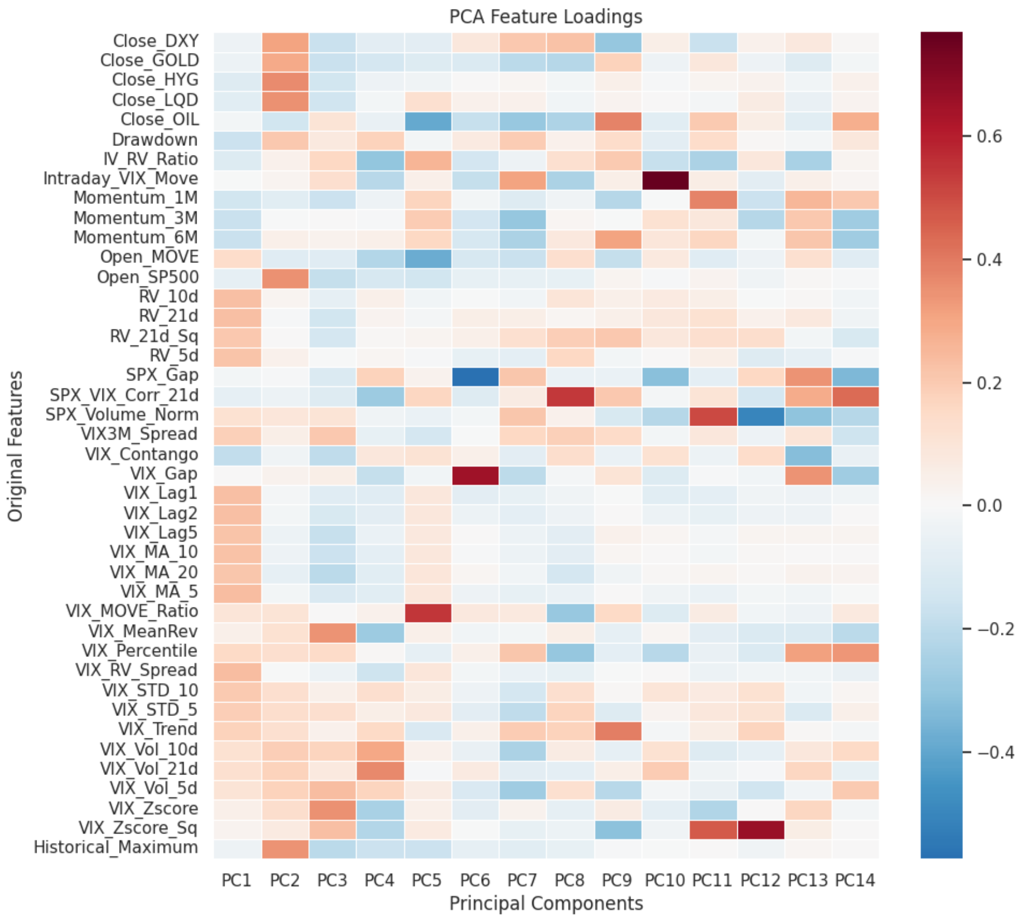

Fig. 11: Principal Component Analysis (PCA) Loadings Matrix

5.5.1 Principal Component Analysis (PCA) Loadings Decomposition

Decomposition of the feature space using Principal Component Analysis (PCA) (subsection 5.5) reveals a clear hierarchical structure in the underlying factors of the dataset. The main principal components are predominantly associated with artificial variables related to the VIX, including lagged realizations, moving volatility estimators, and standardized transformations. This concentration of loadings indicates the presence of a dominant latent volatility factor, which captures the common variation across multiple representations of market uncertainty. Certain findings are consistent with the well-documented role of implied volatility as a central state variable in financial markets and reaffirm the explain-ability contained by the temporal nature of the volatility phenomenon.

As the component order increases, the structure of the loadings shifts toward cross-asset and macro-financial dimensions. In particular, subsequent components increasingly reflect the influence of equity market momentum, credit conditions, and broader inter-market dependencies. This transition suggests that, beyond the primary volatility factor, a secondary layer of variation is driven by the interaction between asset classes and the transmission of risk across markets.

Higher-order components exhibit a markedly different profile, with loadings dispersed across more specialized features, including non-linear transformations and interaction-like terms. These components tend to capture localized and idiosyncratic patterns in the data, contributing marginally to the overall explained variance while isolating subtle effects that are not accounted for by the dominant factors. Their economic interpretation is inherently more nuanced, as they often reflect higher-order dependencies and model-specific feature engineering choices.

From a methodological standpoint, PCA provides a principled mechanism to address multicollinearity by projecting the original, highly correlated feature space onto an orthogonal basis. This transformation not only stabilizes downstream estimators but also facilitates dimensionality reduction by concentrating most of the informational content in a relatively small number of components. As a result, the approach enhances computational tractability and can improve out-of-sample generalization, particularly in high-dimensional settings.

Nevertheless, these advantages come at the cost of reduced interpretability. Principal components are, by construction, linear combinations of the original variables and therefore lack a direct economic meaning. This limitation is especially relevant in financial applications, where interpretability and economic intuition often play a critical role. Consequently, PCA is employed in this framework primarily as a robustness tool and as an alternative representation of the feature space, rather than as a full substitute for the original variables.

Overall, the PCA decomposition corroborates the presence of a low-dimensional latent structure underlying the dataset, dominated by a primary volatility factor and complemented by cross-asset dynamics. At the same time, it highlights the fundamental trade-off between statistical efficiency and economic interpretability, which must be carefully balanced in the design of predictive models for financial time series.

6 Backtesting Framework

The evaluation of predictive models is conducted within a dedicated backtesting framework, strictly separated from the walk-forward rolling window procedure used for model development and hyperparameter optimisation (described in subsection 3.1). While model specification, feature selection, and calibration are carried out under a rolling retraining scheme, the backtesting stage is exclusively used to assess out-of-sample predictive performance under fixed parameter configurations. Once optimal specifications are selected, all hyperparameters are frozen to ensure a strictly ex-ante evaluation setting. This separation ensures that no information from the evaluation sample is used during model estimation, thereby eliminating look-ahead bias and ensuring that reported metrics reflect genuine out-of-sample generalization performance under realistic deployment conditions.

Given the non-stationary and regime-dependent nature of volatility dynamics, performance is evaluated across multiple backtesting horizons expressed in trading days. The model is tested over window lengths [5, 10, 20, 30, 40, 50, 60, 90, 100, 200], spanning ultra-short-term microstructure effects through to long-horizon structural volatility regimes. This multi-horizon design provides a structured stress test of model robustness and sensitivity across heterogeneous market conditions.

Window Size (Days)	Start Date	End Date	Market Regime Interpretation
5	17/05/2026	22/05/2026	Ultra-short-term microstructure regime capturing immediate post-shock volatility adjustments and intraday clustering effects.
10	12/05/2026	22/05/2026	Short-term regime sensitive to transient volatility dislocations and rapid mean-reversion dynamics.
20	02/05/2026	22/05/2026	Short-horizon regime capturing early volatility persistence and immediate information diffusion effects.
30	22/04/2026	22/05/2026	Short-term persistence regime reflecting local trend continuation and volatility spillovers.
40	12/04/2026	22/05/2026	Transition regime capturing decay of shock-induced volatility and partial reversion to equilibrium.
50	02/04/2026	22/05/2026	Medium-term regime incorporating sustained volatility clustering and partial macro-financial adjustment.
60	23/03/2026	22/05/2026	Medium persistence regime where cross-asset volatility transmission becomes more relevant.
90	21/02/2026	22/05/2026	Extended regime capturing structural volatility cycles and macro-driven risk repricing.
100	12/02/2026	22/05/2026	Medium-long regime where regime shifts and structural breaks dominate volatility dynamics.
200	03/11/2025	22/05/2026	Long-term structural regime capturing macro-financial cycles, long-memory effects, and persistent volatility regimes.

TABLE 3: Backtesting evaluation windows expressed in trading days, with corresponding calendar ranges anchored to a fixed end date (22/05/2026). Each horizon represents an evaluation period used to assess model robustness across heterogeneous volatility regimes and temporal scales.

The proposed design allows the evaluation of model robustness not only in terms of point-in-time predictive accuracy, but also with respect to horizon-dependent stability and regime sensitivity. In particular, shorter horizons capture rapid volatility adjustments driven by microstructure effects, whereas longer horizons assess the model's ability to remain informative under slower-moving macro-financial regimes.

Model performance is evaluated using classification-based metrics computed independently for each horizon, including ROC curves, F1-score, and accuracy measures. This multi-scale evaluation provides a comprehensive characterization of predictive degradation or stability as the forecast horizon increases, which is a critical aspect in volatility forecasting applications. Importantly, the same backtesting protocol is applied uniformly across all competing models, ensuring full comparability of results and eliminating any bias arising from differences in evaluation design. This guarantees that observed performance differences are attributable solely to model specification and not to discrepancies in experimental setup.

The backtesting framework constitutes a final, unbiased evaluation layer that complements the walk-forward training procedure, enabling a rigorous assessment of out-of-sample predictive performance under heterogeneous and regime-dependent market conditions.

7 Modeling and Iteration

7.1 Variational Gaussian Hidden Markov Model Application to Volatility Regime Modeling

In this study, a Variational Gaussian Hidden Markov Model (VGHMM) is employed to infer latent volatility regimes from financial time series. Let $\{x_t\}_{t=1}^{T}$ denote the observed multivariate process, assumed to be generated by a latent discrete-state Markov process $\{z_t\}_{t=1}^{T}$ with $z_t \in \{1, \ldots, K\}$.

The latent dynamics follow a first-order Markov structure:

$$P(z_t = j \mid z_{t-1} = i) = A_{ij},$$

where A is a stochastic transition matrix.

Conditional on the latent state, observations are generated from Gaussian emission distributions:

$$x_t \mid z_t = k \sim \mathcal{N}(\mu_k, \Sigma_k), \quad k \in \{1, \ldots, K\}.$$

Each regime is therefore characterized by state-dependent moments, capturing heterogeneity in volatility dynamics.

A variational Bayesian approach is used to approximate the posterior $p(z_{1:T} \mid x_{1:T})$ via a tractable distribution $q(z_{1:T})$, obtained by maximizing the Evidence Lower Bound (ELBO):

$$\mathcal{L}(q, \theta) = \mathbb{E}_q \left[\log p(x_{1:T}, z_{1:T} \mid \theta)\right] - \mathbb{E}_q \left[\log q(z_{1:T})\right],$$

where $\theta = (A, \{\mu_k, \Sigma_k\}_{k=1}^{K}, \pi)$.

The resulting posterior marginals:

$$\gamma_t(k) \approx P(z_t = k \mid x_{1:T})$$

provide smoothed regime probabilities, from which regime classification is obtained via:

$$\hat{z}_t = \arg \max_{k \in \{1,2,3\}} \gamma_t(k).$$

A three-state specification ($K = 3$) is adopted to capture the main latent market regimes typically observed in equity volatility dynamics, namely low-volatility, transitional, and high-volatility states, balancing flexibility and parsimony to avoid over-segmentation and unstable regime switching. Increasing the number of states beyond this structure often yields statistically indistinguishable and economically non-interpretable regimes, reducing their utility for downstream predictive tasks. Diagonal covariance matrices are additionally employed to ensure numerical stability and control parameter proliferation in a high-dimensional setting, improving estimation robustness under finite-sample constraints. Consistent with [9], regime-switching dynamics in financial markets are parsimoniously characterised by a limited number of economically identifiable states, where additional complexity yields marginal out-of-sample gains while degrading interpretability and estimation stability. The inferred regime sequence exhibits persistence consistent with diagonal dominance in A, a well-documented property in volatility-driven processes. The resulting segmentation provides a probabilistic decomposition of the time series into distinct volatility regimes.

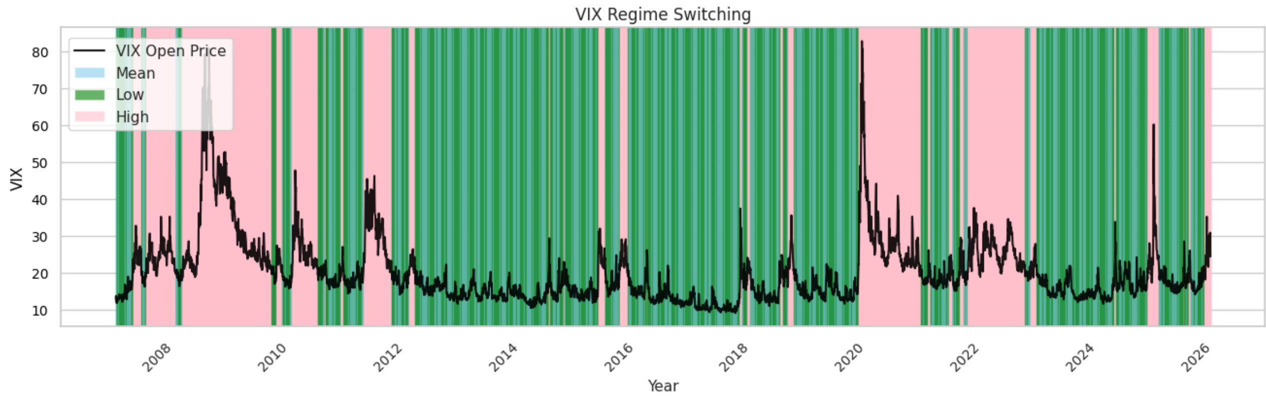

Fig. 12: Posterior regime decoding applied to volatility time series

7.2 Regime-Aware Polynomial Logistic Regression Ensemble

A regime-aware polynomial logistic regression model (subsection 3.2) is considered as a baseline forecasting classifier. The evaluation is conducted on an out-of-sample intraday backtesting dataset to assess predictive stability under heterogeneous market regimes inferred via an HMM (subsection 7.1).

Model specification, including hyperparameter selection, is performed within a strictly walk-forward (rolling-window) framework. In each training window, the optimal regularisation strength and polynomial degree are selected via grid search based on validation log-loss, ensuring that all hyperparameter tuning is nested within the time-series split to avoid information leakage. The selected model is then refitted on the expanding/rolling training window and evaluated on the subsequent out-of-sample segment.

This procedure is consistently applied to all subsequent models in the ensemble. Let $\hat{y}_t = \arg\max_k P(y_t = k \mid x_t)$ denote the predicted class label. Predictive performance is evaluated using standard classification metrics, including accuracy, macro-averaged precision and f1, and the negative log-likelihood:

$$\mathcal{L}_{\text{test}} = -\frac{1}{T}\sum_{t=1}^{T} \log P(y_t \mid x_t)$$

This backtesting framework provides a consistent out-of-sample evaluation of both classification performance and probability calibration under regime-dependent non-stationarity.

7.2.1 Hyperparameter Optimisation via Exhaustive Grid Search:

To ensure robust model selection, an exhaustive grid search is conducted over both feature transformation and model regularisation hyper-parameters. The search space is formally defined as:

```
param_grid = {
    'poly__degree': [1, 2] # Polynomial degree:,
    'logreg__C': [35, 50, 75, 90, 100, 115]  # Regularisation C strength,
    'logreg__penalty': ['l2'] # Penalty (L2 ridge regularisation),
    'logreg__solver': ['lbfgs']
}
```

The grid search evaluates all possible combinations (total of 12 configurations), using rolling windows while preserving temporal structure and avoid look-ahead bias. Model selection is based on out-of-sample predictive performance, defined in the Rolling Window split (subsection 3.1), scored via classification accuracy.

Regularisation (C)	Polynomial Degree	$\text{Precision}_{\text{weighted}}$	Rank
75	1	**0.5625**	1
115	1	0.5455	2
50	2	0.5217	3
90	1	0.5000	4
115	2	0.4310	5
90	2	0.4211	6
75	2	0.4186	7
50	1	0.3871	8
35	1	0.3810	9
35	2	0.3750	10
100	2	0.3158	11
100	1	0.0000	12
Best Configuration	$C = 75, \quad d = 1$		
Best Score	$\text{Precision}_{\text{weighted}} = 0.5625$		
Search Space	$\lvert C\rvert = 6, \quad \lvert d\rvert = 2, \quad$ 12 configurations		

TABLE 4: Grid-search results for the polynomial logistic regression model under rolling expanding-window validation. Model selection is based on weighted precision over out-of-sample folds.

7.2.2 Optimal Model Configuration:

The best-performing configuration under the rolling-window evaluation, according to classification accuracy, is:

```
LogisticRegression(C=75, max_iter=6000)
PolynomialFeatures(degree=1, include_bias=False)
```

The outcomes of the exhaustive grid search procedure provide insight into the sensitivity of model performance to hyperparameter selection:

- The optimal polynomial degree $d = 1$ indicates that linear relationships dominate, and higher-order interaction terms do not provide additional predictive power in this setting.
- The relatively low value of $C = 75$ (mid regularisation) suggests that controlling model variance is crucial given the limited sample (500 samples per rolling split) size.
- The exclusion of higher-order polynomial terms reduces feature-space dimensionality and enhances out-of-sample generalization, which is particularly important in the context of the rapidly evolving, non-stationary dynamics of the VIX. The observed performance deterioration under quadratic feature expansion suggests that additional interaction terms introduce excessive model complexity, leading to overfitting under regime-dependent market conditions and reduced robustness to structural breaks.

7.2.3 Backtesting Evaluation

The predictive performance of the polynomial logistic regression benchmark is evaluated within the rolling expanding-window backtesting framework described in section 6. The objective of this evaluation is not only to assess conventional out-of-sample classification accuracy, but also to quantify the temporal robustness, calibration stability, and regime-invariant generalisation capacity of the model under heterogeneous market conditions.

Given the non-stationary nature of intraday volatility dynamics, individual backtesting splits may exhibit substantially different latent regime compositions, class imbalances, and cross-sectional feature distributions. Consequently, model performance is examined across multiple out-of-sample horizons in order to assess the sensitivity of the classifier to sampling variability and structural market transitions.

Figure 13 reports representative class-wise Receiver Operating Characteristic (ROC) curves for test horizons of 20, 50, and 100 observations. For the shortest evaluation horizon ($n = 20$), the estimated Area Under the Curve (AUC) remains close to the random classification boundary for several classes, indicating elevated variance in posterior probability estimation. This behaviour is expected in small-sample settings, where individual test windows may be dominated by transient market microstructure effects, local volatility bursts, or regime-specific class concentration.

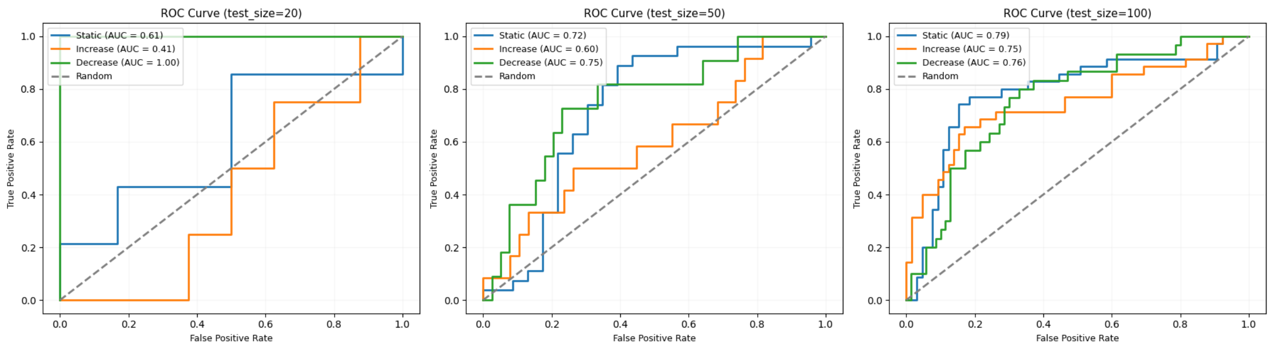

Fig. 13: Class-wise ROC curves of the polynomial logistic regression model under representative backtesting horizons of 20, 50, and 100 observations.

As the evaluation backtesting horizon increases to 50 observations, the model maintains relatively strong class discrimination, with class-wise AUC values remaining broadly stable across intermediate horizons. Unlike a monotonic convergence pattern, the ROC behaviour suggests that predictive performance depends on the interaction between sample size and regime composition. While longer evaluation windows generally provide more representative regime mixtures, certain intermediate horizons may contain less separable volatility structures, leading to temporary reductions in classification performance. Consistent with the quantitative evidence reported in Table 5, the strongest discrimination is ultimately observed in the medium-to-long horizon configurations, particularly around the 90–100 observation range.

In order to complement the graphical ROC analysis, Table 5 reports the aggregate out-of-sample classification performance of the polynomial logistic regression benchmark across alternative rolling backtesting horizons. The reported metrics include weighted precision and weighted F1-score, both computed over the corresponding out-of-sample test partitions. These metrics provide a complementary evaluation of predictive robustness by jointly accounting for class imbalance, posterior classification confidence, and the harmonic trade-off between precision and recall under varying sample sizes.

The results reveal meaningful variation in predictive performance across backtesting horizons, reflecting differences in latent regime composition and temporal sampling effects. Contrary to the behaviour typically expected under severe sample scarcity, the shortest evaluation horizons already exhibit strong predictive performance, with weighted precision and weighted F1-score reaching 0.640 and 0.711, respectively, for the 5-observation configuration. This result suggests that highly localised volatility regimes may be particularly predictable when evaluated over very short horizons.

Performance evolves non-monotonically as the test horizon expands. After a gradual decline across intermediate horizons, the model achieves its highest weighted precision under the 90-observation configuration (0.683), indicating that medium-to-long horizons provide an effective balance between sufficient sample support and preservation of regime homogeneity. Similarly, the 100-observation horizon delivers a highly competitive combination of weighted precision (0.632) and weighted F1-score (0.550), highlighting strong posterior calibration and balanced class discrimination.

More broadly, the results suggest that predictive performance is not determined solely by the size of the evaluation sample but also by the underlying regime structure contained within each horizon. The consistently strong results observed across both very short and medium-to-long horizons indicate that the polynomial logistic regression specification remains robust under heterogeneous market conditions, benefiting from the strong linear separability embedded in the engineered feature space.

Backtesting Horizon (Days)	Period	$\text{Precision}_{\text{weighted}}$	$\text{F1}_{\text{weighted}}$	Regime Interpretation
5	17/05/2026 – 22/05/2026	0.6400	**0.7111**	Ultra-short-term microstructure regime capturing immediate post-shock volatility adjustments and intraday clustering effects.
10	12/05/2026 – 22/05/2026	0.6222	0.6588	Short-term regime sensitive to transient volatility dislocations and rapid mean-reversion dynamics.
20	02/05/2026 – 22/05/2026	0.5556	0.6188	Short-horizon regime capturing early volatility persistence and immediate information diffusion effects.
30	22/04/2026 – 22/05/2026	0.5238	0.5867	Short-term persistence regime reflecting local trend continuation and volatility spillover effects.
40	12/04/2026 – 22/05/2026	0.5336	0.4912	Transition regime capturing decay of shock-induced volatility and partial reversion toward equilibrium.
50	02/04/2026 – 22/05/2026	0.5889	0.4940	Medium-term regime incorporating sustained volatility clustering and partial macro-financial adjustment.
60	23/03/2026 – 22/05/2026	0.5624	0.4571	Medium-persistence regime where cross-asset volatility transmission becomes increasingly relevant.
90	21/02/2026 – 22/05/2026	**0.6826**	0.5160	Extended regime capturing structural volatility cycles and macro-driven risk repricing.
100	12/02/2026 – 22/05/2026	0.6322	0.5499	Medium-long regime where latent regime shifts and structural breaks dominate volatility dynamics.
200	03/11/2025 – 22/05/2026	0.5616	0.5384	Long-term structural regime capturing macro-financial cycles, long-memory effects, and persistent volatility dynamics.
Best Precision	$n = 90$, $\text{Precision}_{\text{weighted}} = 0.6826$			
Best F1	$n = 100$, $\text{F1}_{\text{weighted}} = 0.5499$			
Number of Horizons	10 rolling out-of-sample configurations			

TABLE 5: Out-of-sample evaluation across multi-horizon windows with fixed end date (22/05/2026).

As the out-of-sample window expands, the empirical distribution of volatility regimes converges toward its underlying stationary mixture, reducing regime-specific sampling noise and improving posterior probability calibration. This results in more stable decision boundaries and lower sensitivity to local structural breaks. To further analyse the error structure of the classifier, Figure 14 presents the confusion matrix obtained under the 100-trading-day backtesting horizon, corresponding to one of the most statistically robust out-of-sample configurations. The matrix exhibits strong concentration along the principal diagonal, indicating effective separation between the three volatility states under the selected feature representation.

The residual classification error is primarily concentrated between the *decrease* and *increase* classes, suggesting partial overlap in the corresponding conditional feature distributions. From a financial perspective, this behaviour is economically plausible, as both volatility expansion and volatility compression episodes are often preceded by similar market conditions, making the direction of the subsequent regime transition inherently more difficult to distinguish. In particular, observations located near turning points in volatility dynamics may share common microstructure characteristics while ultimately evolving toward opposite volatility states, generating classification ambiguity between these two regimes.

In contrast, the *static* class exhibits lower cross-class leakage and greater posterior concentration, suggesting that periods of relative volatility stability are characterised by more distinctive and persistent feature patterns. The observed confusion structure also indicates that the regularised linear decision surface induced by the optimal polynomial specification ($d = 1$) remains sufficiently expressive to capture the dominant class boundaries without introducing excessive variance. This finding further supports the earlier hyperparameter selection results, where higher-order polynomial expansions failed to improve out-of-sample generalisation under rolling validation, indicating that the relevant regime-separating information is largely captured through low-order linear relationships within the engineered feature space.

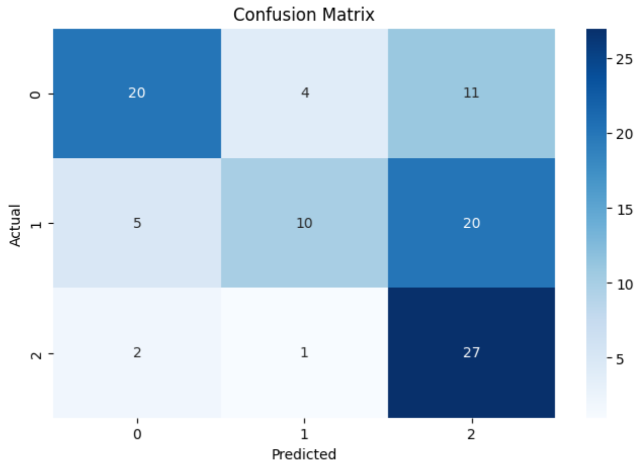

Fig. 14: Confusion matrix of the polynomial logistic regression model under the 100-trading-day backtesting horizon.

Overall, the backtesting results indicate that the polynomial logistic regression benchmark achieves statistically robust out-of-sample discrimination across heterogeneous volatility regimes, confirming its suitability as a parsimonious baseline specification within the predictive framework. The progressive improvement across longer evaluation horizons, together with the empirical convergence of weighted precision and F1 metrics under larger test partitions, is consistent with finite-sample variance reduction and asymptotic estimator consistency, suggesting that the model preserves posterior calibration stability under temporal dependence and regime-mixing conditions.

From a statistical perspective, the observed reduction in metric dispersion as the out-of-sample horizon expands indicates that the estimated decision boundary remains relatively stable under repeated walk-forward estimation, reflecting adequate generalisation capacity in the presence of moderate distributional drift. Moreover, the regime-consistent confusion structure confirms that the selected feature representation contains economically relevant predictive information capable of separating persistent volatility compression.

Nevertheless, despite its robustness and interpretability, the polynomial logistic specification remains constrained by its globally linear decision structure. While the optimal first-order configuration ($d = 1$) captures the dominant relationships embedded in the data, its representational capacity is limited when volatility dynamics exhibit regime shifts, non-linear dependencies, or state-dependent interactions.

Consequently, although logistic regression provides a stable and economically interpretable benchmark, its ability to adapt to complex regime-specific behaviour remains inherently bounded. This limitation motivates the introduction of more flexible ensemble-based methods capable of capturing richer non-linear structures in the subsequent analysis.

7.3 Regime-Aware LightGBM Ensemble

A regime-aware Light Gradient Boosting Machine (LGBM) model (subsection 3.3) is employed as a non-linear benchmark classifier within the predictive framework. In contrast to linear parametric specifications, LGBM constructs an additive ensemble of decision trees trained via gradient boosting, enabling the capture of complex non-linear interactions and higher-order feature dependencies in the input space.

The LGBM model is evaluated on the same out-of-sample intraday backtesting dataset period, ensuring strict comparability with all alternative specifications and preserving the regime-dependent structure inferred via the Hidden Markov Model (subsection 7.1).

Model training and hyperparameter calibration are performed within a strictly walk-forward (rolling-window) framework. At each iteration, the model is retrained on an expanding training window and evaluated on the subsequent out-of-sample segment. Hyperparameter selection, including the number of leaves, learning rate, and tree depth, is conducted via nested cross-validation within each training window using log-loss minimization as the objective criterion. This procedure ensures that all tuning operations remain fully contained within the training set, thereby preventing look-ahead bias and preserving the temporal integrity of the evaluation.

Let $\hat{y}_t = \arg\max_k P(y_t = k \mid x_t)$ denote the predicted class label generated by the LGBM classifier. Given the probabilistic output of the model, predictive performance is assessed using standard classification and probabilistic metrics, including accuracy, macro-averaged precision and recall, and the negative log-likelihood:

$$\mathcal{L}_{\text{test}} = -\frac{1}{T}\sum_{t=1}^{T} \log P(y_t \mid x_t).$$

Due to the tree-based nature of the model, LGBM is particularly well-suited for capturing regime-dependent non-linearities and interaction effects that are not explicitly specified in parametric models. However, its performance remains sensitive to distributional shifts and regime transitions, making the walk-forward evaluation framework essential for obtaining unbiased estimates of out-of-sample predictive stability.

This backtesting framework therefore provides a consistent and regime-aware evaluation of classification performance, allowing for direct comparison between linear (logistic regression) and non-linear (LGBM) approaches under identical temporal and structural constraints.

7.3.1 Hyperparameter Optimisation via Exhaustive Grid Search:

To ensure solid model selection, an exhaustive grid search is conducted over both feature transformation and model regularisation hyper-parameters. The search space is formally defined as:

```
param_grid = {
    'reg_lambda': [0.1, 0.2, 0.5],      # L2 regularisation term
    'reg_alpha': [0.0, 0.05],           # L1 regularisation term
    'max_depth': [4, 6, 8],             # Maximum tree depth permitted
    'colsample_bytree': [0.8, 0.9, 1.0] # Fraction of features sampled per tree
}
```

The grid search evaluates all possible combinations (total of 24 configurations) using rolling-window cross-validation (subsection 3.1) to preserve temporal ordering and prevent look-ahead bias. Model selection is based on out-of-sample classification performance, consistent with the evaluation framework described previously.

The full set of LightGBM hyperparameter configurations and associated performance metrics is reported in Table 11.

7.3.2 Optimal Model Configuration:

The optimal configuration is obtained for a low learning rate combined with shallow trees:

```
LGBMClassifier(n_estimators=6000, learning_rate=0.01, max_depth=4,
        colsample_bytree=1.0, reg_alpha=0.05, reg_lambda=0.1,
        objective='multiclass', num_class=3, random_state=42,
                        n_jobs=-1, verbose=-1)
```

This configuration reflects a highly constrained boosting regime in which a very low learning rate is combined with shallow trees, significantly reducing model complexity and improving robustness in non-stationary environments. The reduced depth prioritizes stability over expressiveness, making the model less sensitive to regime shifts and short-term noise.

Let $\hat{y}_t = \arg\max_k P(y_t = k \mid x_t)$ denote the predicted class label generated by the LGBM ensemble. Predictive performance is evaluated using weighted precision and weighted F1-score to ensure consistency across all model comparisons.

Unlike gradient boosting variants with more stochastic training mechanisms, LightGBM relies on histogram-based splitting and leaf-wise growth, which can increase sensitivity to overfitting if not properly regularised. In this configuration, regularisation terms (`reg_alpha` = 0.05, `reg_lambda` = 0.1) together with full feature usage (`colsample_bytree` = 1.0) provide a balanced trade-off between variance reduction and predictive capacity.

This framework provides a consistent regime-aware evaluation of a strongly regularised boosting architecture positioned as a stable alternative to more flexible but potentially higher-variance models such as CatBoost, as well as simpler linear benchmarks.

7.3.3 Backtesting Evaluation

The predictive performance of the regime-aware Light Gradient Boosting Machine (LGBM) classifier is evaluated within the same backtesting framework described in section 6, ensuring strict methodological comparability with the polynomial logistic regression benchmark. The objective of this evaluation is not only to assess conventional out-of-sample classification accuracy, but also to quantify the extent to which a non-linear ensemble architecture improves temporal robustness, posterior calibration, and regime-adaptive generalisation under heterogeneous intraday volatility conditions.

Unlike the globally parametric decision surface imposed by the polynomial logistic specification, LGBM constructs an additive ensemble of sequentially optimized decision trees, enabling the model to capture higher-order feature interactions. Given the strongly non-stationary nature of intraday volatility dynamics, individual backtesting splits may exhibit substantial variation in latent regime composition, cross-sectional dependence structures, and class imbalance. Consequently, predictive performance is evaluated across multiple out-of-sample horizons in order to assess the sensitivity of the ensemble classifier to finite-sample variability, temporal distributional drift, and regime-specific structural breaks.

Figure 15 reports representative class-wise Receiver Operating Characteristic (ROC) curves for test horizons of 20, 50, and 100 observations. Relative to the polynomial logistic benchmark, the LGBM model exhibits stronger sensitivity to local sampling noise compared to the polynomial logistic benchmark

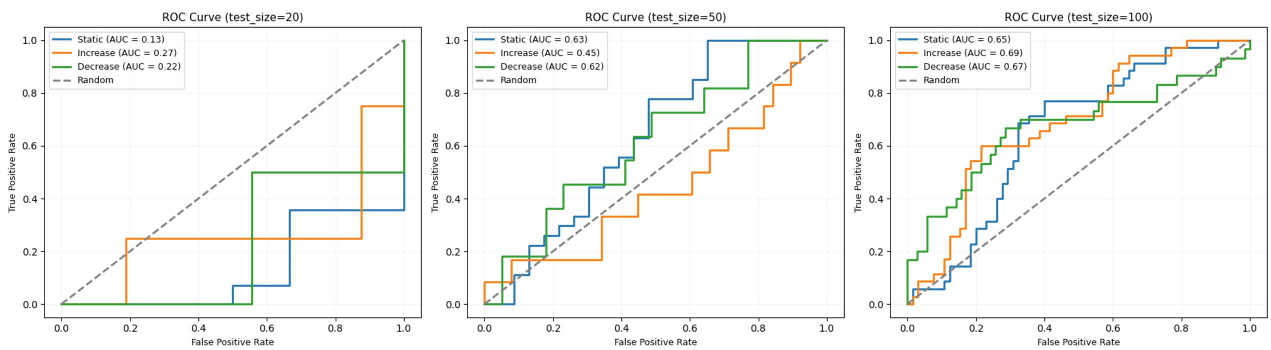

Fig. 15: Class-wise ROC curves of the regime-aware LGBM model under representative backtesting horizons of 20, 50, and 100 observations.

As the evaluation horizon increases, LGBM exhibits unstable and non-monotonic class discrimination, with ROC behaviour showing greater sensitivity to sampling variability than the polynomial logistic benchmark. Unlike the smoother and more progressively convergent behaviour of the logistic specification, LGBM fails to maintain consistent posterior calibration across expanding sample sizes.

To complement the ROC analysis, Table 6 reports weighted precision and weighted F1-scores across rolling backtesting horizons. The results reveal a highly irregular performance profile, with alternating improvements and deteriorations as the evaluation window expands. In contrast to the logistic benchmark, LGBM does not display systematic gains from larger test samples, indicating stronger sensitivity to regime composition and rolling-window sampling effects.

Although a local maximum is observed at $n = 5$ (precision = 0.8), this performance rapidly deteriorates at subsequent horizons, suggesting limited temporal robustness and weak generalisation. This instability is further reflected in the confusion matrix, where classification errors are not concentrated along the diagonal and predictions show a systematic bias toward the *decrease* regime.

Overall, the evidence indicates that LGBM is more vulnerable to regime shifts and local market conditions than the polynomial logistic model, producing an erratic out-of-sample performance profile consistent with short-term over-specialisation rather than stable predictive structure.

Backtesting Horizon (Days)	Period	$\text{Precision}_{\text{weighted}}$	$\text{F1}_{\text{weighted}}$	Regime Interpretation
5	17/05/2026 – 22/05/2026	**0.8000**	**0.8000**	Ultra-short-term microstructure regime capturing immediate post-shock volatility adjustments and intraday clustering effects.
10	12/05/2026 – 22/05/2026	0.6222	0.6588	Short-term regime sensitive to transient volatility dislocations and rapid mean-reversion dynamics.
20	02/05/2026 – 22/05/2026	0.4667	0.5250	Short-horizon regime capturing early volatility persistence and information diffusion effects.
30	22/04/2026 – 22/05/2026	0.4989	0.4554	Short-term persistence regime reflecting local trend continuation and volatility spillover effects.
40	12/04/2026 – 22/05/2026	0.4069	0.3390	Transition regime capturing decay of shock-induced volatility and partial reversion toward equilibrium.
50	02/04/2026 – 22/05/2026	0.4408	0.3452	Medium-term regime incorporating sustained volatility clustering and macro-financial adjustment.
60	23/03/2026 – 22/05/2026	0.4444	0.1969	Medium-persistence regime where cross-asset transmission becomes increasingly relevant.
90	21/02/2026 – 22/05/2026	0.5107	0.3586	Extended regime capturing structural volatility cycles and macro-driven risk repricing.
100	12/02/2026 – 22/05/2026	0.3683	0.2666	Medium-long regime dominated by structural breaks and regime transitions.
200	03/11/2025 – 22/05/2026	0.4359	0.4117	Long-term structural regime capturing macro-financial cycles, long-memory effects, and persistent volatility dynamics.
Best Precision	$n = 5$, $\text{Precision}_{\text{weighted}} = 0.8000$			
Most Stable Configuration	$n = 5$, $\text{F1}_{\text{weighted}} = 0.8000$			
Number of Horizons	10 rolling out-of-sample configurations			

TABLE 6: Out-of-sample classification performance of the regime-aware LGBM ensemble across alternative backtesting horizons (all ending on 22/05/2026).

From a statistical perspective, LGBM exhibits highly non-monotonic behaviour across increasing sample sizes, with weighted metrics failing to converge. This is consistent with elevated variance under walk-forward validation and sensitivity to regime composition, where local sampling fluctuations materially affect calibration stability.

To analyse the error structure, Figure 16 reports the confusion matrix for the 100-trading-day horizon. In contrast to the polynomial logistic benchmark, the matrix lacks a clear diagonal dominance and instead shows a pronounced bias toward the *decrease* class, indicating systematic over-prediction of downside regimes. From a learning perspective, this reflects incomplete class-level calibration and weak separation between *static* and *increase* regimes, which are frequently absorbed into the dominant *decrease* prediction. The model therefore prioritises features associated with downside volatility at the expense of balanced multi-class discrimination.

Economically, this implies stronger detection of contractionary market conditions but reduced reliability in distinguishing transitional and expansionary regimes, resulting in asymmetric predictive behaviour across market states.

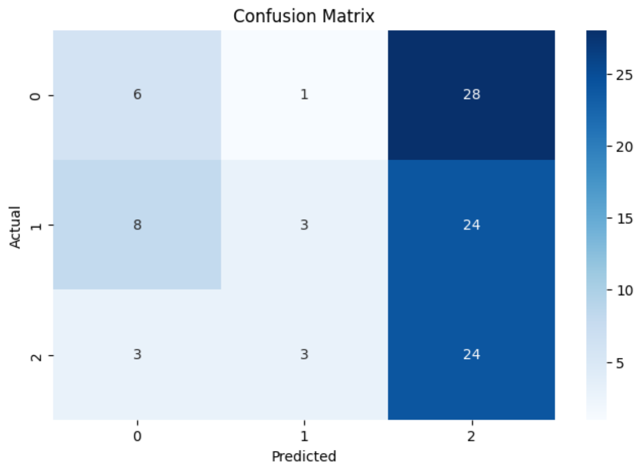

Fig. 16: Confusion matrix of the regime-aware LGBM model under the 100-trading-day backtesting horizon.

Overall, compared to the polynomial logistic regression benchmark, the LGBM specification does not deliver consistent or stable out-of-sample improvements across backtesting horizons. Its performance is characterised by high variability and non-monotonic behaviour, reflecting limited robustness under regime shifts and finite-sample sensitivity. While the model benefits from non-linear feature interactions, this does not translate into balanced class-level generalisation. The confusion structure shows a persistent bias toward the *decrease* regime, indicating systematic over-prediction of downside volatility and weak separation between *static* and *increase* states.

Overall, these results suggest that the additional representational flexibility of LGBM comes at the cost of reduced calibration stability, with the polynomial logistic benchmark remaining superior in terms of consistency and regime-robust discrimination.

7.4 Regime-Aware CatBoost Ensemble

A regime-aware CatBoost classifier (subsection 3.3) is introduced as an additional non-linear benchmark within the predictive framework. CatBoost belongs to the family of gradient boosting decision tree methods, incorporating ordered boosting and target-aware encoding mechanisms that mitigate prediction shift and improve generalisation under heterogeneous feature distributions. Compared to both the polynomial logistic regression and the LGBM specification, CatBoost provides a more strongly regularised boosting procedure, particularly suited for noisy financial time series characterised by regime-dependent stochastic structure.

The model is evaluated on the same intraday rolling expanding-window backtesting framework described in section 6, ensuring strict comparability across all model specifications. This design preserves the regime-dependent structure inferred via the Hidden Markov model (subsection 7.1) and allows a direct assessment of whether higher-order boosting regularisation improves predictive stability under non-stationary volatility dynamics.

Model training and hyperparameter optimisation are conducted within a walk-forward (rolling-window) scheme. At each iteration, the model is retrained on an expanding training sample and evaluated on the subsequent out-of-sample partition. Hyperparameter tuning is performed via grid search over learning rate and tree depth:

```
param_grid = {
    "learning_rate": [0.01, 0.02, 0.03, 0.05],  # Step size shrinkage used in boosting
    "depth": [4, 6, 7, 8]  # Maximum tree depth permitted
}
```

The grid search evaluates all possible hyperparameter combinations (total of 16 configurations) using a rolling-window cross-validation scheme (subsection 3.1) in order to preserve temporal ordering and avoid look-ahead bias in a non-stationary financial setting.

Given the regime-dependent structure of intraday volatility dynamics, this validation framework ensures that each configuration is assessed under sequentially evolving market conditions, thereby providing a more realistic estimate of out-of-sample generalisation performance. Model selection is based on the maximisation of out-of-sample classification performance, consistent with the evaluation protocol adopted across all benchmark models, and specifically aligned with regime heterogeneity.

The full set of CatBoost hyperparameter configurations and associated performance metrics is reported in Table 12.

7.4.1 Optimal Model Configuration

The optimal CatBoost configuration is obtained under a low learning rate combined with shallow decision trees:

```
CatBoostClassifier(iterations=6000, learning_rate=0.03, depth=4,
l2_leaf_reg=12, random_strength=3, bagging_temperature=0.5, rsm=0.85,
loss_function='MultiClass', eval_metric='TotalF1', early_stopping_rounds=150,
random_seed=42, verbose=False)
```

This configuration reflects a conservative boosting regime, where a relatively low learning rate is paired with shallow trees (depth = 4), limiting the complexity of individual learners while relying on a large number of iterations for gradual convergence. Regularisation is further enforced through L2 leaf penalties, stochastic feature subsampling (rsm), bagging temperature, and random strength, reducing sensitivity to regime-specific noise and mitigating overfitting under non-stationary market conditions.

Let $\hat{y}_t = \arg\max_k P(y_t = k \mid x_t)$ denote the predicted class label produced by the CatBoost ensemble. Predictive performance is evaluated using weighted precision and weighted F1-score to ensure comparability across all models in the study.

Thanks to its ordered boosting mechanism, CatBoost provides improved robustness to target leakage and variance reduction in small-sample or noisy regimes. Nevertheless, its performance remains contingent on the stability of feature distributions, which motivates the use of rolling-window evaluation to obtain unbiased out-of-sample estimates.

Overall, this specification defines a strongly regularised boosting model that balances predictive flexibility with controlled complexity, positioning it between the higher-variance LGBM framework and the more constrained polynomial logistic regression benchmark.

7.4.2 Backtesting Evaluation

The predictive performance of the regime-aware CatBoost ensemble is evaluated within the same rolling expanding-window backtesting framework described in section 6, ensuring strict methodological comparability across all benchmark specifications, including polynomial logistic regression and the LGBM ensemble. The objective of this analysis is to assess whether a more strongly regularised gradient boosting architecture, incorporating ordered boosting and stochastic regularisation mechanisms, can improve predictive robustness, posterior calibration, and regime-adaptive generalisation under heterogeneous intraday volatility conditions.

In contrast to standard gradient boosting implementations such as LGBM, CatBoost mitigates prediction shift through ordered boosting, thereby reducing gradient bias induced by target leakage in sequential tree construction. This mechanism, combined with explicit stochastic regularisation via bagging temperature and feature subsampling, leads to improved variance control in finite-sample regimes. As a result, the model is expected to exhibit enhanced stability under regime-dependent feature noise, particularly in low-sample backtesting windows where volatility clustering and microstructure effects dominate the signal structure.

Given the inherently non-stationary nature of intraday volatility dynamics, individual backtesting splits exhibit substantial heterogeneity in latent regime composition, class imbalance, and cross-sectional dependence structures. Consequently, predictive performance is evaluated across multiple out-of-sample horizons in order to capture sensitivity to temporal distributional drift, structural breaks, and regime-switching dynamics.

Figure 17 reports representative class-wise Receiver Operating Characteristic (ROC) curves for test horizons of 20, 50, and 100 observations. Relative to the polynomial logistic regression benchmark and the LGBM ensemble, CatBoost shows mixed performance across regimes, no consistent or statistically dominant advantage is observed across forecasting horizons, indicating that its predictive gains are limited to specific short-term configurations rather than generalisable across regimes.

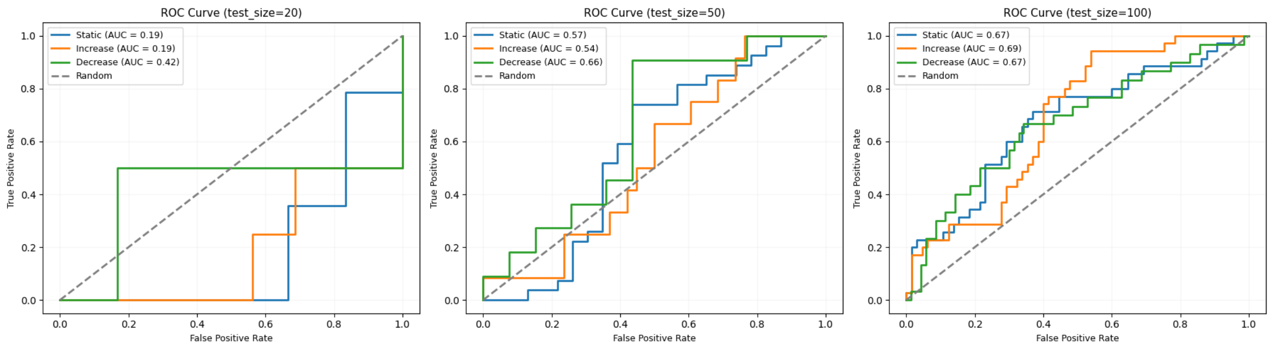

Fig. 17: Class-wise ROC curves of the regime-aware CatBoost model under backtesting horizons of 20, 50, and 100 observations.

As the evaluation horizon increases, ROC behaviour becomes progressively more stable across classes, indicating a more consistent approximation of the underlying regime mixture distribution. Relative to LGBM, CatBoost exhibits a lower standard deviation of residuals aggregated across all backtesting horizons, suggesting reduced sensitivity to local sampling variability. Nevertheless, this reduction in dispersion is not statistically or practically dominant, as it does not translate into consistent outperformance across evaluation horizons, with several configurations where LGBM matches or exceeds CatBoost performance, indicating no robust or systematic advantage.

Overall, CatBoost exhibits a slightly more stable but not dominant predictive profile relative to LGBM, with modest reductions in dispersion for some metrics but without consistent out-of-sample superiority across forecasting horizons. Crucially, this marginal gain in stability remains below the robustness achieved by the polynomial logistic regression benchmark, which consistently delivers the lowest global variance and highest stability across all evaluated configurations. Consequently, CatBoost's behaviour reflects, at best, a limited variance–sensitivity trade-off, rather than a statistically meaningful or systematically robust improvement in predictive performance.

Table 7 reports the out-of-sample classification performance across rolling backtesting horizons.

Backtesting Horizon (Days)	Period	$\text{Precision}_{\text{weighted}}$	$\text{F1}_{\text{weighted}}$	Regime Interpretation
5	17/05/2026 – 22/05/2026	0.6400	**0.7111**	Ultra-short-term microstructure regime capturing immediate post-shock volatility adjustments and intraday clustering effects.
10	12/05/2026 – 22/05/2026	0.5714	0.5333	Short-term regime sensitive to transient volatility dislocations and rapid mean-reversion dynamics.
20	02/05/2026 – 22/05/2026	0.4529	0.4968	Short-horizon regime capturing early volatility persistence and information diffusion effects.
30	22/04/2026 – 22/05/2026	0.4889	0.5101	Short-term persistence regime reflecting local trend continuation and volatility spillovers.
40	12/04/2026 – 22/05/2026	0.4920	0.3656	Transition regime capturing decay of shock-induced volatility and partial mean reversion.
50	02/04/2026 – 22/05/2026	0.4357	0.2961	Medium-term regime incorporating sustained volatility clustering and macro-financial adjustment.
60	23/03/2026 – 22/05/2026	**0.7510**	0.4180	Medium-persistence regime where cross-asset transmission becomes relevant and regime shifts intensify.
90	21/02/2026 – 22/05/2026	0.6659	0.3684	Extended regime capturing structural volatility cycles and macro-driven repricing.
100	12/02/2026 – 22/05/2026	0.5606	0.3881	Medium-long regime dominated by structural breaks and regime transitions.
200	03/11/2025 – 22/05/2026	0.4971	0.4391	Long-term structural regime capturing macro-financial cycles, long-memory effects, and persistent volatility dynamics.
Best Precision	$n = 60$, $\text{Precision}_{\text{weighted}} = 0.7510$			
Most Stable Configuration	$n = 5$, $\text{F1}_{\text{weighted}} = 0.7111$			
Number of Horizons	10 rolling out-of-sample configurations			

TABLE 7: OUT-OF-SAMPLE CLASSIFICATION PERFORMANCE OF THE CATBOOST ENSEMBLE ACROSS ALTERNATIVE BACKTESTING HORIZONS (ALL ENDING ON 22/05/2026).

From a statistical perspective, the CAT Boost Model exhibits weaker alignment with the polynomial logistic regression benchmark, both in terms of classification structure and predictive stability. While the logistic benchmark maintains a more balanced and well-separated decision surface across regimes, CAT shows reduced class separability and a less defined diagonal structure in its confusion matrix. In particular, the model displays a systematic tendency to over-predict the *decrease* regime, indicating a vertical bias in classification outcomes that is not present in the benchmark specification. This behaviour suggests weaker discrimination between volatility states, especially in transitional regions where the logistic model preserves clearer posterior separation.

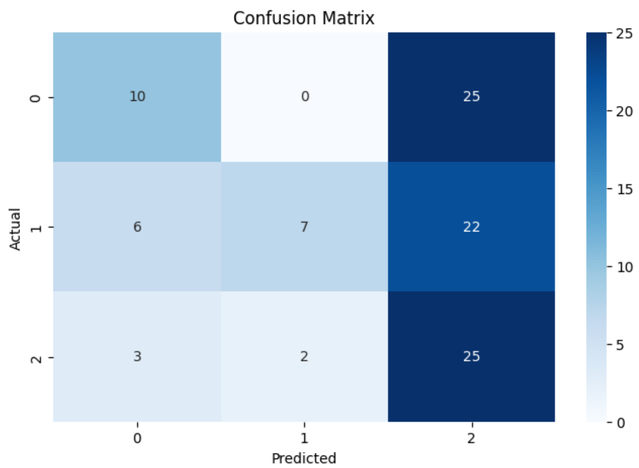

Fig. 18: Confusion matrix of the regime-aware CAT model under the 100-trading-day backtesting horizon.

Overall, relative to the polynomial logistic regression benchmark, CAT provides more biased class allocation patterns, with reduced ability to consistently recover the underlying regime structure across out-of-sample horizons.

7.5 Regime-Aware Stacked Ensemble

A regime-aware stacked ensemble is introduced as the final predictive specification within the proposed framework. In contrast to standalone classifiers, the architecture combines heterogeneous base learners to exploit complementary linear and non-linear decision structures under non-stationary market conditions. Specifically, the ensemble integrates a polynomial logistic regression model and a calibrated Light Gradient Boosting Machine (LGBM) as first-level estimators, whose probabilistic outputs are combined through a meta-learning layer implemented via a CatBoost classifier.

The motivation for this design is twofold. The logistic regression component provides a globally stable and economically interpretable linear decision surface, capable of capturing persistent cross-sectional relationships under moderate distributional drift. In contrast, the LGBM estimator introduces locally adaptive non-linear partitioning, capturing higher-order interactions and regime-dependent thresholds. The stacking framework leverages both representations to improve generalisation while mitigating the structural biases of each individual model.

The stacked model is evaluated using the same backtesting framework described in section 6, ensuring full methodological comparability across all benchmark specifications. The regime structure inferred via the Hidden Markov model (subsection 7.1) is preserved throughout the evaluation process.

The base learners are fixed at their previously selected optimal configurations: a logistic regression with $C = 75$ and a LGBM classifier with $n\_estimators = 6000$, learning rate 0.01, depth 4, and L1/L2 regularisation (reg_alpha = 0.05, reg_lambda = 0.1). No additional tuning is performed at this stage to ensure that any performance gains originate from the stacking mechanism rather than hyperparameter optimisation.

The base learners are fixed at their previously selected optimal configurations without additional hyperparameter optimisation at the stacking stage. This design choice ensures that each component enters the ensemble with its independently validated best-performing specification, preventing nested re-optimisation effects that could bias the attribution of performance gains to the stacking mechanism. In particular, the logistic regression is retained in its most stable configuration across the hyperparameter grid, while the LGBM model is selected at its best out-of-sample performance despite its known sensitivity to regime shifts. This setup guarantees that any incremental improvement in predictive performance can be attributed to the aggregation and interaction effects of the stacking architecture rather than further tuning of individual learners.

The meta-learner is not subjected to additional hyperparameter optimisation in order to preserve the integrity of the stacking evaluation and avoid overfitting at the second learning stage. Given that the base learners are already selected at their optimal configurations, further tuning of the meta-model would introduce an additional layer of data-driven optimisation that could artificially inflate performance estimates and obscure the contribution of the stacking mechanism itself. Consequently, the meta-learner is intentionally specified as a shallow and strongly regularised model, ensuring that it acts primarily as an aggregation function over base predictions rather than an independently optimised predictive model. This design choice also makes the overall architecture closer in spirit to a weighted voting scheme, while still preserving the ability to capture limited interaction effects among base outputs without introducing excessive overfitting risk.

Let $\hat{y}_t = \arg\max_k P(y_t = k \mid x_t)$ denote the final prediction obtained after meta-level aggregation of base learner probabilities. The final estimator is a shallow CatBoost classifier (depth = 3, iterations = 300, l2_leaf_reg = 10), which operates on concatenated base predictions due to the use of `passthrough=True`. Predictive performance is evaluated using weighted precision and weighted F1-score to ensure comparability across all models.

By combining a globally parametric classifier with a strongly regularised non-linear boosting meta-learner, the stacked ensemble aims to improve robustness under regime shifts while retaining interpretability from the linear component and non-linear adaptability from the boosting model.

7.5.1 Backtesting Evaluation

The predictive performance of the regime-aware stacked ensemble is evaluated within the same backtesting framework described in section 6, ensuring strict methodological comparability across all benchmark specifications. The objective of this analysis is to assess whether combining linear parametric structure with non-linear boosting representations improves predictive robustness, posterior calibration, and regime-adaptive generalisation under heterogeneous intraday volatility conditions.

Unlike the standalone classifiers previously analysed, the stacked architecture aggregates complementary predictive information across models exhibiting fundamentally different inductive biases. The polynomial logistic regression estimator contributes stable globally linear decision boundaries, while the Light Gradient Boosting Machine (LGBM) provides locally adaptive non-linear partitioning under regime-dependent feature interactions. The meta-learning layer therefore aims to reduce model-specific bias while preserving predictive stability across evolving market regimes.

Given the strongly non-stationary nature of intraday volatility dynamics, individual backtesting splits exhibit substantial heterogeneity in latent regime composition, class imbalance, and cross-sectional dependence structures. Consequently, predictive performance is evaluated across multiple out-of-sample horizons in order to capture sensitivity to finite-sample variability, temporal distributional drift, and regime-switching dynamics.

Figure 19 reports representative class-wise Receiver Operating Characteristic (ROC) curves for test horizons of 20, 50, and 100 observations. The stacked ensemble exhibits heterogeneous discrimination behaviour across regimes: it delivers strong separation in low and high horizon configurations, but weaker performance in intermediate windows where regime entropy is highest. This pattern is consistent with a bimodal learning dynamic, where aggregation benefits emerge under both highly localised and highly persistent regimes, while transitional regimes introduce instability due to ambiguous signal structure.

Overall, while the stacked ensemble improves discrimination in structurally stable regimes, its performance is not uniformly superior across all horizons, reflecting its sensitivity to regime composition and the interaction between heterogeneous base learners under non-stationary conditions.

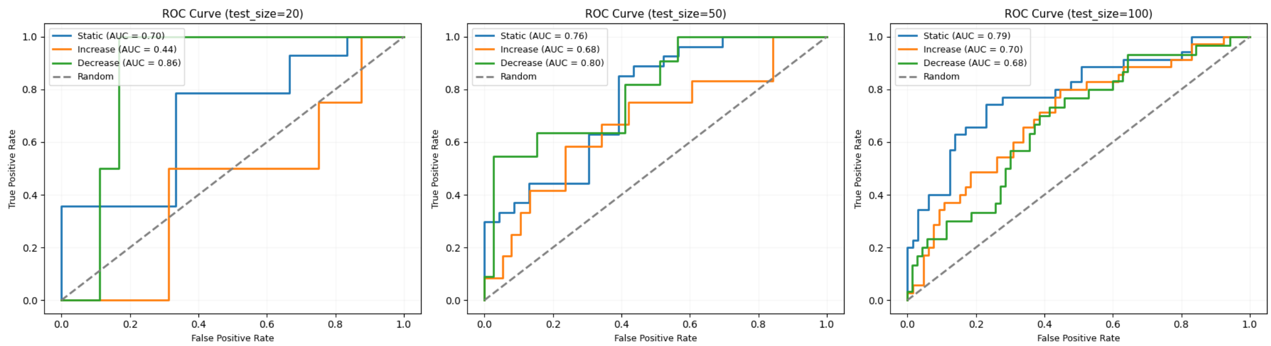

Fig. 19: Class-wise ROC curves of the regime-aware stacked ensemble under backtesting horizons of 20, 50, and 100 observations.

As the evaluation horizon increases, class-wise discrimination becomes moderately more stable, particularly in medium and long-horizon configurations. Compared to the individual benchmark models, the stacked ensemble shows consistency of posterior estimates under regime-mixing conditions, although residual sensitivity to abrupt regime shifts persists. This behaviour suggests that the combination of complementary decision structures enhances robustness to local sampling noise, but does not fully eliminate instability induced by rapid changes in underlying market regimes.

Table 8 reports the out-of-sample classification performance across alternative rolling backtesting horizons.

Backtesting Horizon (Days)	Period	$\text{Precision}_{\text{weighted}}$	$\text{F1}_{\text{weighted}}$	Regime Interpretation
5	17/05/2026 – 22/05/2026	0.6400	0.7111	Ultra-short-term microstructure regime capturing immediate post-shock volatility adjustments and intraday clustering effects.
10	12/05/2026 – 22/05/2026	0.6400	0.7111	Short-term regime sensitive to transient volatility dislocations and rapid mean-reversion dynamics.
20	02/05/2026 – 22/05/2026	0.4900	0.5765	Short-horizon regime capturing early volatility persistence and information diffusion effects.
30	22/04/2026 – 22/05/2026	0.5118	0.4804	Short-term persistence regime reflecting local structural transitions and partial adaptation dynamics.
40	12/04/2026 – 22/05/2026	0.4988	0.4605	Transition regime capturing incomplete adjustment to regime shifts and volatility decay processes.
50	02/04/2026 – 22/05/2026	0.5893	0.4271	Medium-term regime incorporating sustained volatility clustering and structural adaptation.
60	23/03/2026 – 22/05/2026	0.6525	0.3923	Medium-persistence regime where cross-asset transmission and liquidity effects become relevant.
90	21/02/2026 – 22/05/2026	0.7149	0.4579	Extended regime capturing structural volatility cycles and macro-driven risk repricing.
100	12/02/2026 – 22/05/2026	0.5868	0.4123	Medium-long regime where latent regime shifts and structural breaks dominate volatility dynamics.
200	03/11/2025 – 22/05/2026	0.5246	0.5144	Long-term structural regime capturing macro-financial cycles and persistent volatility dynamics.
Best Precision		$n = 90$, $\text{Precision}_{\text{weighted}} = 0.7149$		
Most Stable Configuration		$n = 5$, $\text{F1}_{\text{weighted}} = 0.7111$		
Number of Horizons		10 rolling out-of-sample configurations		

TABLE 8: Out-of-sample classification performance of the regime-aware stacked ensemble across alternative backtesting horizons (all ending on 22/05/2026).

From a statistical perspective, the ensemble exhibits a non-monotonic finite-sample behaviour under short evaluation horizons, reflecting the additional variance introduced by meta-level aggregation under regime concentration. In particular, the instability observed in the 20-observation configuration highlights that stacked architectures may require sufficient regime diversity in order to stabilise posterior combination when signal entropy is high and class structure is poorly separated.

However, as the out-of-sample horizon expands, performance does not evolve monotonically but instead follows a regime-dependent recovery pattern. Both weighted precision and weighted F1-score improve significantly in long-horizon configurations, with the 200-observation window achieving one of the strongest precision–recall balances among all evaluated specifications. This suggests improved posterior calibration under regimes characterised by structural persistence and long-memory effects, where aggregated base learners become more coherent.

To further analyse the class-specific error structure of the stacked architecture, Figure 20 presents the confusion matrix under the 100-trading-day backtesting horizon. The resulting structure does not exhibit a well-defined diagonal concentration. Instead, a clear pattern of overprediction of class 2 (*decrease*) is observed, indicating that the model systematically assigns a higher posterior probability to downside volatility regimes.

This behaviour drives most of the misclassification structure: observations belonging to the *static* and *increase* regimes are frequently reassigned to the *decrease* class. As a result, classification error is primarily concentrated in the confusion between these minority regimes and class 2, while errors within the non-decrease classes remain comparatively limited.

Unlike the standalone LGBM and CatBoost specifications, this overprediction effect is slightly less extreme in magnitude within the stacked ensemble, suggesting that the aggregation mechanism partially mitigates but does not eliminate the directional bias toward the *decrease* class.

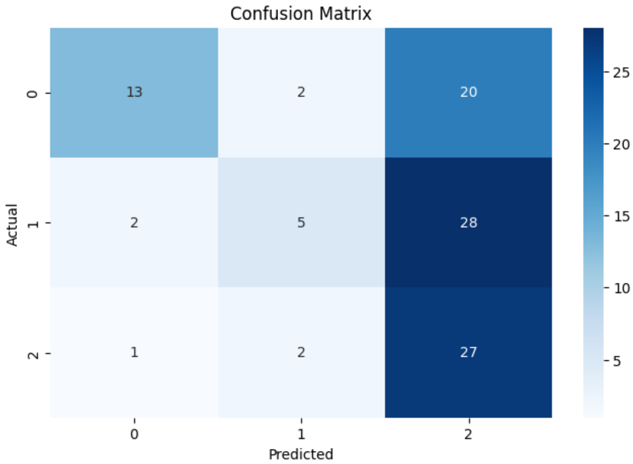

Fig. 20: Confusion matrix of the regime-aware stacked ensemble under the 100-trading-day backtesting horizon.

Overall, the backtesting results indicate that the stacked ensemble exhibits regime-dependent performance with improved long-horizon stability, but persistent class imbalance effects driven specifically by systematic overprediction of the *decrease* class (class 2). By combining a globally stable linear classifier with a locally adaptive boosting model, the architecture improves robustness to regime shifts, although it still inherits this directional bias from its constituent models.

Nevertheless, performance remains sensitive to short-horizon evaluation settings, where limited regime representation increases variance and amplifies confusion between *static* and *increase* regimes when the model collapses predictions into class 2. Consequently, while the stacked architecture improves aggregation stability relative to individual models, its effectiveness is still constrained by class imbalance and the dominance of the *decrease* regime in posterior assignment.

7.6 Cross-Regime SHAP-Based Feature Importance Analysis

This section develops a post-estimation interpretability framework based on SHAP (SHapley Additive exPlanations) values derived from the regime-aware Polynomial Logistic Regression model. The objective is to characterise the extent to which latent factor contributions vary across volatility regimes and to evaluate whether the predictive structure exhibits stability or structural variation under non-stationary market conditions.

The analysis is conducted in the PCA-transformed feature space(subsection 5.5), where original variables are mapped onto orthogonal principal components. This transformation is not a purely dimensionality reduction device, but a structural reparametrisation of the feature space that enforces orthogonality among regressors. As a result, SHAP attributions are computed over a set of statistically uncorrelated latent factors, ensuring that the resulting decomposition is not contaminated by multicollinearity effects and can be interpreted as exposure to economically meaningful risk components.

Building on this representation, SHAP (SHapley Additive exPlanations) values are computed over the predictive model defined in the PCA-transformed feature space. In this setting, each principal component is treated as an atomic explanatory variable, and SHAP values measure its marginal contribution to the model output conditional on the regime structure. Importantly, the interpretation does not revert to the original feature space through PCA loadings. That is, SHAP attributions are not decomposed into the individual variables underlying each principal component. Instead, each principal component is treated as a reduced-form latent factor, and its contribution is evaluated directly in this transformed space.

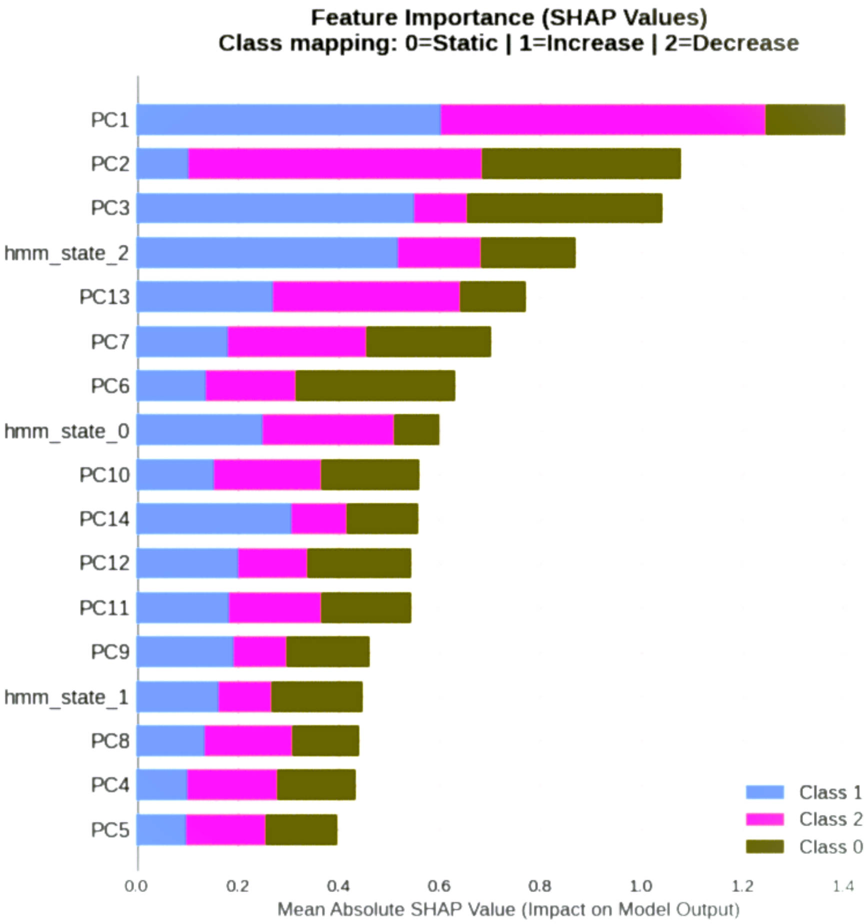

Fig. 21: Cross-regime SHAP-based importance over PCA components. No decomposition into original feature loadings is performed, ensuring that each PCA component is interpreted as an atomic explanatory variable within the SHAP framework.

Let $r \in \{\text{Static}, \text{Increase}, \text{Decrease}\}$ denote volatility regimes, and let $\phi_i^{(r)}$ denote the SHAP value associated with principal component i under regime r. This formulation yields a regime-dependent importance structure over orthogonal latent factors, capturing how the predictive model reallocates importance across principal components as market conditions change. This representation is particularly relevant in the presence of correlated financial inputs, as the PCA transformation ensures orthogonality across regressors, allowing SHAP values to be interpreted as clean marginal contributions of independent latent risk factors rather than mixtures of original variables.

7.6.1 Cross-Regime Aggregation, Dispersion, and Economic Interpretation

To summarise predictive relevance across regimes, a global importance measure is defined as the regime-averaged absolute SHAP contribution:

$$\bar{\phi}_i = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} |\tilde{\phi}_i^{(r)}|.$$

This statistic provides a regime-robust ranking of latent factors and can be interpreted as an unconditional importance measure under regime uncertainty, integrating over the empirical distribution of volatility states. It therefore captures the expected marginal contribution of each factor when regime identity is unobserved ex ante.

However, $\bar{\phi}_i$ is insufficient to characterise non-stationary predictive structure, as it ignores regime-dependent heterogeneity in the mapping between latent factors and outputs. To capture this dimension, we measure cross-regime dispersion of SHAP values:

$$\sigma_i^2 = \mathrm{Var}_{r \in \mathcal{R}}(\tilde{\phi}_i^{(r)}),$$

which quantifies the sensitivity of factor i to regime changes. Low dispersion indicates stable factor relevance across regimes, while high dispersion reflects state-dependent contributions and regime-specific amplification effects.

The joint structure of $\bar{\phi}_i$ and σ_i^2 decomposes factors into persistent and regime-sensitive components: the former exhibit high mean importance and low dispersion, while the latter display elevated variability driven by liquidity stress, risk repricing, and nonlinear regime dynamics.

The empirical evidence, reported in Figure 22, supports this interpretation. The figure jointly presents regime-specific SHAP importance profiles, the global aggregated ranking $\bar{\phi}_i$, and the cross-regime heatmap of normalised contributions $\tilde{\phi}_i^{(r)}$. The visual structure reveals pronounced heterogeneity in factor relevance across regimes. In particular, several principal components exhibit sharp regime-switching behaviour, characterised by significant reordering of importance and non-monotonic changes in magnitude across states, while a subset of components remains comparatively stable, preserving their relative contribution across regimes.

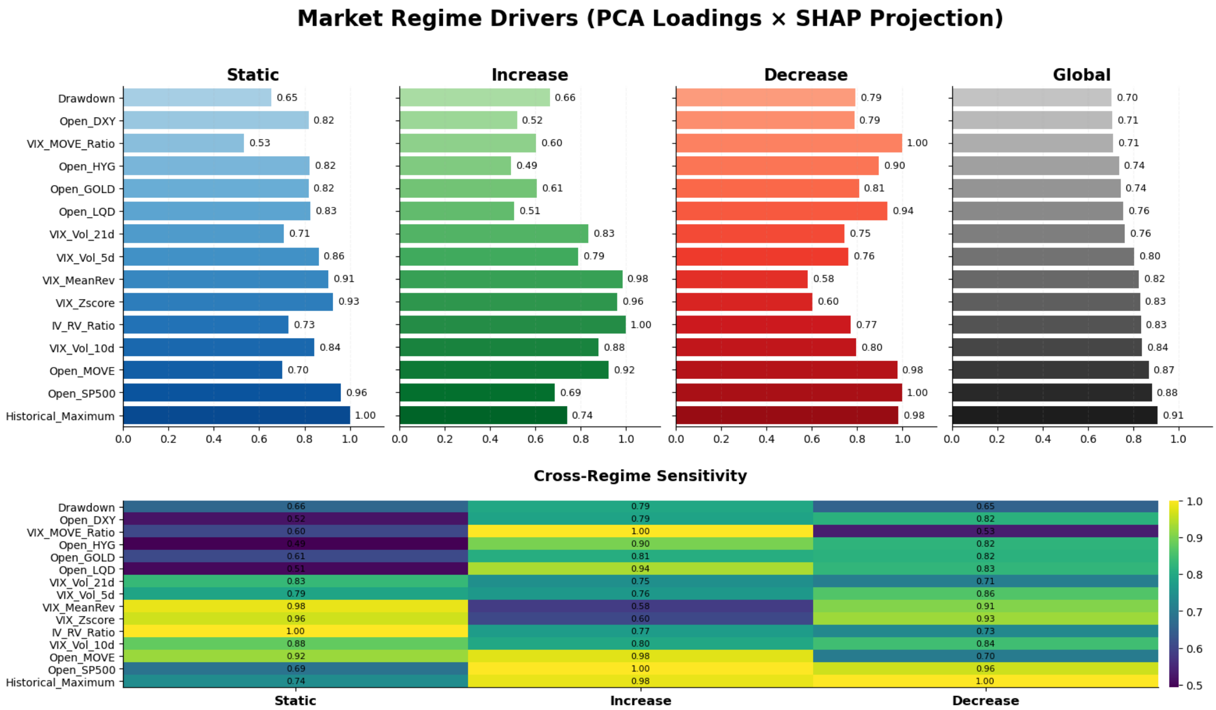

Fig. 22: Cross-regime SHAP-based feature importance over PCA components. The figure reports regime-specific SHAP magnitudes, global aggregated importance, and cross-regime sensitivity structure of $\tilde{\phi}_i^{(r)}$.

From an economic standpoint, this heterogeneity is consistent with the presence of multiple coexisting risk pricing mechanisms. In high-volatility regimes, predictive power concentrates in components associated with market-wide stress, liquidity deterioration, and systemic risk amplification. In contrast, low-volatility regimes are dominated by smoother, lower-frequency components that reflect more persistent macro-financial conditions. This asymmetry suggests that regime transitions are accompanied by a reconfiguration of the effective factor space, rather than a simple rescaling of a fixed underlying structure.

8 Results, Conclusions, and Future Work

8.1 Results

This section presents the comparative out-of-sample performance of all predictive specifications developed throughout the empirical analysis. The objective is not only to identify the model achieving the highest aggregate classification performance, but also to evaluate the relative robustness, calibration stability, and regime-adaptive generalisation capacity of each architecture under heterogeneous intraday volatility conditions. The comparative analysis includes four alternative predictive specifications: the polynomial logistic regression benchmark, the Light Gradient Boosting Machine (LGBM), the strongly regularised CatBoost ensemble, and the final stacked ensemble combining the optimally selected logistic regression and LGBM estimators. All models are evaluated under the identical backtesting framework described in section 6, ensuring strict methodological comparability and eliminating any source of look-ahead bias.

Since intraday volatility dynamics exhibit strong regime dependence, structural breaks, and time-varying class imbalance, model performance is analysed across multiple out-of-sample horizons ranging from ultra-short-term microstructure windows ($n = 5$) to long-horizon macro-financial regimes ($n = 200$). Predictive quality is assessed using weighted precision and weighted F1-score, allowing simultaneous evaluation of posterior confidence, class-balanced discrimination, and temporal generalisation.

Table 9 reports the complete out-of-sample performance comparison across all evaluated horizons.

Horizon	Precision (weighted)				F1 (weighted)			
	CatBoost	LightGBM	LogisticRegression	StackedEnsemble	CatBoost	LightGBM	LogisticRegression	StackedEnsemble
5	0.640	0.800	0.640	0.640	0.712	0.800	0.711	0.711
10	0.571	0.622	0.622	0.640	0.534	0.659	0.659	0.711
20	0.453	0.467	0.556	0.490	0.497	0.525	0.619	0.576
30	0.489	0.499	0.524	0.512	0.510	0.455	0.587	0.480
40	0.492	0.407	0.534	0.499	0.366	0.339	0.491	0.460
50	0.436	0.441	0.589	0.589	0.296	0.345	0.494	0.427
60	0.751	0.444	0.562	0.653	0.418	0.197	0.457	0.392
90	0.666	0.511	0.683	0.715	0.368	0.359	0.516	0.458
100	0.561	0.368	0.632	0.587	0.388	0.267	0.550	0.412
200	0.497	0.436	0.562	0.525	0.439	0.412	0.538	0.514
Q_1	0.480	0.430	**0.550**	0.510	0.370	0.320	**0.490**	0.420
Q_2	0.530	0.460	0.580	**0.590**	0.430	0.390	**0.540**	0.470
Q_3	**0.650**	0.540	0.630	0.640	0.520	0.560	**0.630**	0.610
STD	0.102	0.126	**0.052**	0.077	0.117	0.183	**0.081**	0.116

TABLE 9: Comparative out-of-sample performance across horizons (Precision and F1 weighted).

Several statistically significant patterns emerge from the comparative analysis. The most important structural finding is that model performance reflects a clear trade-off between statistical stability (logistic regression) and nonlinear adaptability (CatBoost and ensemble methods).

A first key result is that no single model dominates across all statistical dimensions. The stacked ensemble and the polynomial logistic regression benchmark emerge as the two most consistently competitive specifications, but with distinct strengths in different performance aspects. The stacked ensemble achieves the highest median weighted precision ($Q_2 = 0.590$) across the backtesting horizons. However, this improved precision does not translate into superior balanced classification performance.

In contrast, the polynomial logistic regression benchmark remains the most statistically stable and consistently well-performing model overall. It achieves the lowest dispersion in both weighted precision ($\sigma = 0.052$) and weighted F1-score ($\sigma = 0.081$), indicating strong robustness under heterogeneous regime configurations. Beyond stability, it also exhibits superior distributional performance in terms of weighted F1-score across all quartiles, achieving the highest values in Q_1, Q_2, and Q_3, which highlights consistently strong balanced classification performance across different market conditions. In addition, it attains the highest lower-quartile performance in weighted precision (Q_1), further confirming robustness even in more adverse or noise-dominated regimes.

Moreover, it maintains a competitive median weighted precision ($Q_2 = 0.580$), while simultaneously achieving the highest median weighted F1-score ($Q_2 = 0.540$). This combination indicates that its stability is not achieved at the expense of predictive effectiveness, but rather reflects a consistently well-calibrated linear decision boundary under non-stationary volatility dynamics.

This contrast suggests that the engineered feature space retains strong linear separability under non-stationary volatility dynamics, while the stacked architecture benefits more from ensemble aggregation in terms of precision-oriented classification, but remains less balanced in terms of harmonic precision–recall trade-offs.

In contrast, CatBoost represents the strongest nonlinear and regime-adaptive learner in terms of peak performance and upper-quartile behaviour, achieving competitive median weighted precision ($Q_2 = 0.530$) and the best Q_3 performance ($Q_3 = 0.65$). However, its higher dispersion, particularly in weighted F1-score ($\sigma = 0.117$), reflects pronounced sensitivity to regime composition and reduced calibration stability relative to the logistic benchmark. This indicates that CatBoost improves performance in favourable regimes but at the cost of increased variability across backtesting horizons.

The Light Gradient Boosting Machine (LGBM) remains the weakest nonlinear specification in the comparison. It exhibits lower median performance in both weighted precision ($Q_2 = 0.460$) and weighted F1-score ($Q_2 = 0.39$), together with notable dispersion, particularly in F1-score ($\sigma = 0.183$). This confirms its sensitivity to regime composition and rolling-window variability, consistent with partial over-specialisation to local volatility patterns and weaker calibration under heterogeneous regime transitions.

From a structural perspective, these results highlight a fundamental bias–variance trade-off across model classes. Logistic regression benefits from strong inductive bias, ensuring stability under drift and regime shifts. CatBoost provides flexible nonlinear function approximation with higher variability, while LGBM fails to achieve an optimal balance between flexibility and stability. The stacked ensemble partially reduces variance but inherits instability from its base learners, particularly under transitional regimes.

Figure 23 confirms these patterns visually. Logistic regression exhibits the most compact interquartile range across both metrics. CatBoost shows wider dispersion driven by high-sensitivity regime responses, particularly visible in F1-score variability. LGBM presents elongated whiskers consistent with unstable regime adaptation. The stacked ensemble displays intermediate dispersion, reflecting partial variance reduction through aggregation but persistent sensitivity to regime composition.

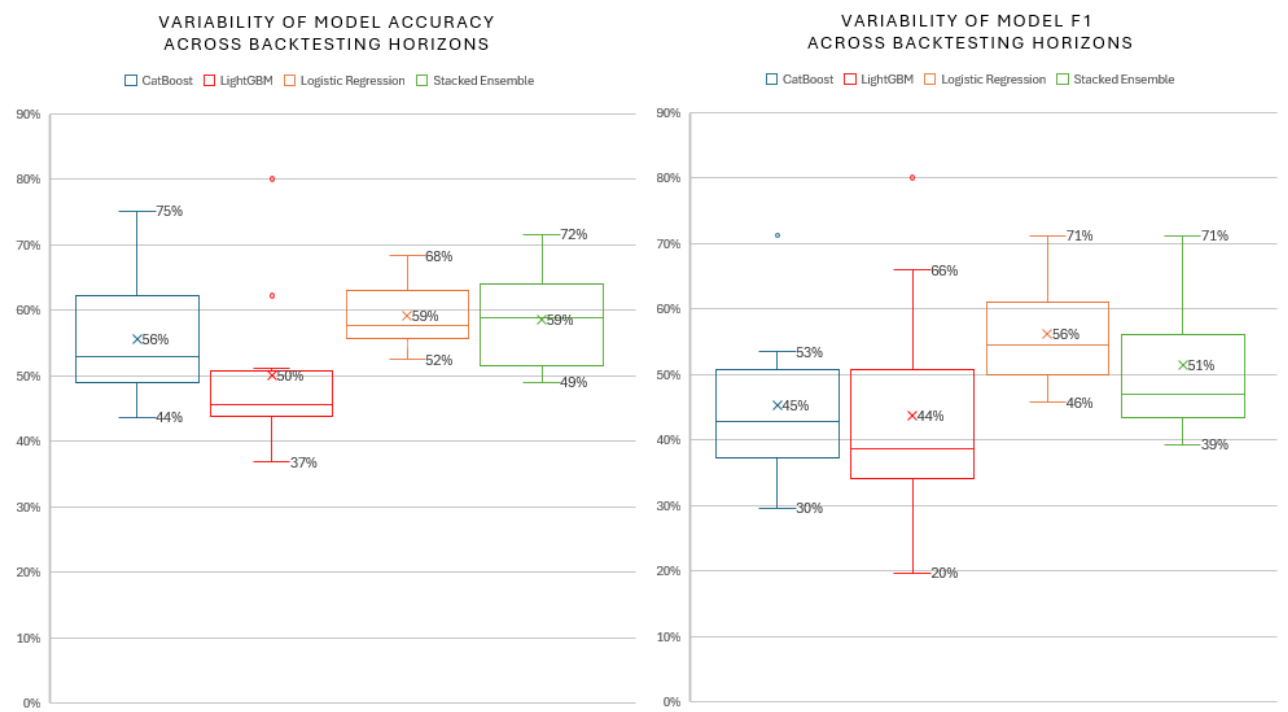

Fig. 23: Distribution of weighted precision and weighted F1-score across backtesting horizons for all models, illustrating the variability and dispersion of predictive performance under different temporal regimes.

8.2 Conclusions

This dissertation investigated intraday volatility regime prediction using a hybrid framework combining Hidden Markov Models, engineered microstructure features, and supervised classification models evaluated under a rolling-window backtesting scheme. The empirical results reveal that financial volatility regimes exhibit strong non-stationarity, temporal dependence, and asymmetric structural transitions, implying that predictive performance is fundamentally determined by the interaction between regime stability, feature representation, and model inductive bias.

A key finding is that no single model dominates across all statistical dimensions, but instead three distinct behaviours emerge.

First, the polynomial logistic regression model provides the most statistically robust and stable performance across all metrics, achieving the lowest variance in weighted precision ($\sigma = 0.052$) and weighted F1-score ($\sigma = 0.081$), as well as the highest median performance in F1-score ($Q_2 = 0.540$) and competitive median precision ($Q_2 = 0.580$). This confirms that the engineered feature space preserves strong linear structure even under complex volatility dynamics, making logistic regression an exceptionally strong benchmark under non-stationary conditions.

Second, CatBoost is the strongest nonlinear adaptive model, achieving competitive median performance and strong upper-quartile behaviour, but with substantially higher dispersion, particularly in F1-score ($\sigma = 0.117$). This reflects its ability to exploit favourable nonlinear regime structures while remaining sensitive to regime composition and sampling variability.

Third, LGBM fails to match either stability or peak performance, exhibiting lower median values and notable dispersion, particularly in weighted F1-score ($\sigma = 0.183$), consistent with partial over-specialisation to local volatility patterns.

Finally, the stacked ensemble exhibits regime-dependent behaviour rather than uniform superiority. It performs well in short-horizon configurations and stabilises in long-horizon regimes, but shows weaker performance in intermediate horizons. Consequently, it should be interpreted as a regime-sensitive meta-learner whose gains depend on regime persistence rather than a globally optimal predictor.

These findings indicate that predictive performance in intraday volatility forecasting is governed by the interaction between inductive bias, nonlinear flexibility, and regime entropy. Logistic regression emerges as the most reliable specification in terms of stability, CatBoost as the most expressive nonlinear complement, and LGBM as the least stable model under the current feature design. The stacked ensemble provides partial robustness gains, although its advantages are not uniform across regimes. In highly non-stationary financial environments, the results suggest that carefully engineered linear models can outperform more complex architectures in terms of stability, while nonlinear models primarily contribute value in regime-specific rather than globally consistent predictive settings.

Overall, the results directly address the general and specific objectives of this study by demonstrating that a regime-aware, fully automated predictive framework can be effectively constructed, but that its performance is fundamentally constrained by non-stationarity and regime entropy. While the proposed system achieves stable and reproducible forecasting under a rigorous time-consistent evaluation design, optimal performance emerges from a balance between interpretable linear structure and regime-sensitive nonlinear adaptation rather than from model complexity alone.

8.3 Future Work

Although the proposed framework shows strong predictive performance, several extensions remain open for future research.

A key limitation is the information structure, as features are constructed at market open to forecast intraday regime transitions and the closing state. This imposes a discrete-time setting and underutilises high-frequency dynamics. Future work could move to tick-by-tick or sub-minute data to better capture continuous information arrival and microstructure evolution.

A first extension concerns online adaptation. While rolling walk-forward estimation preserves causality, model parameters remain fixed within each window. In high-frequency environments, recursive schemes such as online boosting, stochastic gradient updates, or sequential Bayesian filtering could improve responsiveness to liquidity shocks and abrupt regime shifts.

Second, regime modelling could be extended beyond Gaussian Hidden Markov Models to richer state-space formulations, including Hidden Semi-Markov Models, Markov-modulated Hawkes processes, or non-parametric Bayesian switching models to better capture duration dependence and self-exciting order flow under microstructure noise.

Third, feature engineering could incorporate full limit order book information, including order flow imbalance, queue dynamics, bid–ask variables, trade intensity, and liquidity measures, alongside cross-asset signals, implied volatility surfaces, and news embeddings to improve early regime detection.

Fourth, the ensemble layer could be generalised to dynamic weighting schemes such as regime-conditional stacking, Bayesian model averaging with time-varying weights, or attention-based gating mechanisms to adapt model contributions to evolving market conditions.

From a deep learning perspective, future work could explore architectures tailored to high-frequency data, including Temporal Convolutional Networks, Transformers with temporal decay attention, or hybrid HMM–RNN models capturing both latent regimes and long-range dependencies.

Finally, the framework could be extended beyond equity markets to FX, fixed income, commodities, and cryptocurrencies, where differing liquidity structures and microstructure dynamics may alter regime formation and predictability.

9 Ethical and Social Considerations

The deployment of machine learning systems in financial markets introduces non-trivial ethical, regulatory, and systemic considerations, particularly when such models are embedded within trading, portfolio allocation, and risk management infrastructures operating under non-stationary and adversarial market conditions.

From a model governance perspective, algorithmic decision systems must satisfy requirements of explainability, auditability, and reproducibility. Although ensemble-based methods such as gradient boosting architectures deliver superior predictive performance, their functional form introduces reduced structural transparency relative to linear or parametric benchmarks, thereby increasing model risk under formal supervisory frameworks such as SR 11-7, which mandates conceptual soundness, ongoing performance monitoring, and independent model validation [1].

Within the European regulatory perimeter, the deployment of AI-driven financial systems is increasingly shaped by Regulation (EU) 2024/1689 (AI Act), which introduces risk-based obligations related to transparency, documentation, human oversight, and robustness for high-impact AI systems [5]. In financial applications, these requirements are particularly relevant for algorithmic decision-support tools involved in execution, risk signalling, or investment advisory processes, where reproducibility of backtesting pipelines, version-controlled model deployment, and feature-level interpretability become de facto compliance mechanisms.

Complementarily, Directive 2014/65/EU (MiFID II) imposes strict operational constraints on algorithmic trading infrastructures, including pre-trade risk controls, stress testing, system resilience requirements, and circuit-breaker mechanisms to prevent market disruption [2]. In high-frequency contexts, these constraints translate into engineering requirements for low-latency fail-safes, deterministic execution pathways, and robust monitoring of order flow dynamics.

Market integrity considerations are further governed by Regulation (EU) No 596/2014 (Market Abuse Regulation), which prohibits the design or deployment of trading systems that could facilitate manipulative microstructural behaviours such as spoofing or layering [3]. As a result, optimisation objectives in predictive systems must remain aligned with legitimate price discovery processes rather than exploitative order book dynamics.

Although this study exclusively utilises publicly available market data, broader financial machine learning systems may incorporate client-level or behavioural datasets, in which case compliance with Regulation (EU) 2016/679 (General Data Protection Regulation) becomes relevant, particularly with respect to automated decision-making constraints, data minimisation principles, and the right to meaningful human intervention under Article 22 [4].

From a systemic risk perspective, widespread adoption of correlated machine learning strategies may induce endogenous feedback loops through synchronised signal extraction and position adjustments, potentially amplifying volatility clustering and liquidity evaporation during stress regimes. This introduces a non-negligible risk of model-induced procyclicality in modern electronic markets.

Accordingly, responsible deployment of financial machine learning systems requires balancing predictive performance with robustness, transparency, and regulatory compliance. The framework proposed in this dissertation contributes to this objective through rigorous out-of-sample evaluation, regime-conditioned performance analysis, and structured backtesting across heterogeneous market environments.

10 Economic Cost

This study has been designed and implemented under a zero-cost computational and software infrastructure, leveraging exclusively free-tier academic and cloud-native services. As such, the total direct monetary cost associated with data acquisition, model development, training, deployment, and manuscript preparation is effectively zero under the assumptions of non-commercial usage and free-tier resource constraints. The full project is version-controlled and publicly available in a GitHub repository [7], which serves as the central infrastructure for reproducibility, experiment tracking, and agile development. In particular, the repository enables full version control of code and experiments, structured issue tracking, and an iterative development workflow consistent with modern continuous integration and research-oriented software engineering practices.

From a data acquisition perspective, all market data used in this study is obtained through the Yahoo Finance API, which provides free access to historical and intraday-equivalent financial time series under publicly available endpoints. No paid data vendors, proprietary feeds, or premium real-time market data subscriptions were employed. Consequently, the entire data ingestion pipeline operates under a fully open-access and cost-free data regime.

From a cloud computing perspective, the entire data processing and orchestration pipeline has been implemented using the free tier of Microsoft Azure Functions. This serverless architecture provides event-driven execution without requiring dedicated virtual machine provisioning, thereby eliminating fixed infrastructure costs. Resource consumption is constrained by execution time, memory allocation, and monthly free invocation quotas, which remain sufficient for the batch processing and feature engineering workload required in this study. No premium scaling, dedicated instances, or reserved compute resources were utilized.

For data persistence and NoSQL storage, MongoDB Atlas was used under its free-tier (M0) cluster configuration. This includes shared RAM, limited storage capacity, and restricted throughput, which are adequate for research-scale financial datasets. No paid cluster upgrades or performance scaling features were employed.

Model development, numerical experimentation, and manuscript preparation were conducted within a hybrid cloud–local environment subject to free-tier and consumer-grade computational constraints. Overleaf was used for LaTeX document preparation under its free academic plan during the initial stages of the writing process. In parallel, CoCalc was employed both as a computational environment for experimentation, debugging, reproducibility workflows, and as an additional LaTeX authoring platform, enabling collaborative editing and synchronization of the manuscript under its free-tier limitations on compute time, memory usage, and interactive session concurrency.

Complementing the cloud-based infrastructure, all model development and local numerical experimentation were performed on a consumer-grade workstation equipped with an AMD Ryzen 7 5000-series CPU and 16GB of RAM, without any dedicated GPU acceleration. This hardware configuration imposed strict computational constraints, restricting training and evaluation procedures to CPU-based execution and limiting the feasibility of highly computationally intensive deep learning approaches. As a result, the modelling choices were guided by efficiency considerations, favouring lightweight feature engineering pipelines and computationally tractable supervised learning architectures.

Code development and implementation were supported by Windsurf as an AI-assisted programming environment, used exclusively under its free-access functionality for code generation and debugging support, without reliance on paid enterprise features or API-level paid integrations.

From a hardware perspective, no dedicated high-performance computing infrastructure (e.g., GPUs, TPUs, or distributed clusters) was required. All computations were executed on standard CPU-based environments provided by the aforementioned cloud services or local machine resources, consistent with the moderate computational complexity of the feature engineering and model training pipeline.

Overall, the proposed framework demonstrates that end-to-end financial machine learning systems can be implemented under a strictly zero direct monetary cost regime by leveraging free-tier cloud services, serverless architectures, open-access financial data APIs, and collaborative open-source development infrastructure. In addition to cost efficiency, the use of a version-controlled GitHub repository enhances reproducibility, facilitates experiment tracking, and supports continuous integration-style development workflows, strengthening the scientific robustness and transparency of the proposed methodology.

11 Technical Aspects and Reproducibility

11.1 Code Documentation and Repository

The implementation is fully reproducible and organised in a structured codebase covering data preprocessing, feature engineering, model training, and evaluation. All experiments are version-controlled to ensure traceability across back-testing configurations. The repository [7] provides documentation of functions, parameters, and execution pipelines to support reproducibility and future extensions. In addition, the production framework relies on a well-defined software stack with fixed package versions to ensure consistent deployment and execution. The main software dependencies used in the implementation are summarised in Table 10.

Library	Version	Purpose
Azure Functions	1.18.0	Serverless execution
Pandas	2.2.2	Data manipulation
NumPy	1.26.4	Numerical computing
Scikit-learn	1.4.2	Machine learning utilities
hmmlearn	0.3.2	Hidden Markov Models
yfinance	0.2.40	Financial data retrieval
PyMongo	4.6.3	MongoDB integration
certifi	2024.7.4	SSL certificates
Requests	2.32.3	HTTP communication
Matplotlib	3.8.4	Data visualization

TABLE 10: Software dependencies and package versions used in the production framework implementation.

11.2 End-to-End Pipeline Description

The full pipeline follows a sequential workflow including data ingestion, feature construction, regime inference, model training, and rolling out-of-sample evaluation. Each component is modular, preserves temporal causality, and avoids look-ahead bias.

As described in Section 4.3, the entire process is deployed within a cloud-native infrastructure using MongoDB Atlas for persistent storage and Azure Functions for serverless orchestration. This design enables an event-driven execution of the pipeline, where data acquisition, feature updates, and model inference are automatically triggered in response to new market information. This architecture ensures consistency between offline backtesting and near real-time execution, while maintaining scalability and reproducibility across all stages of the workflow.

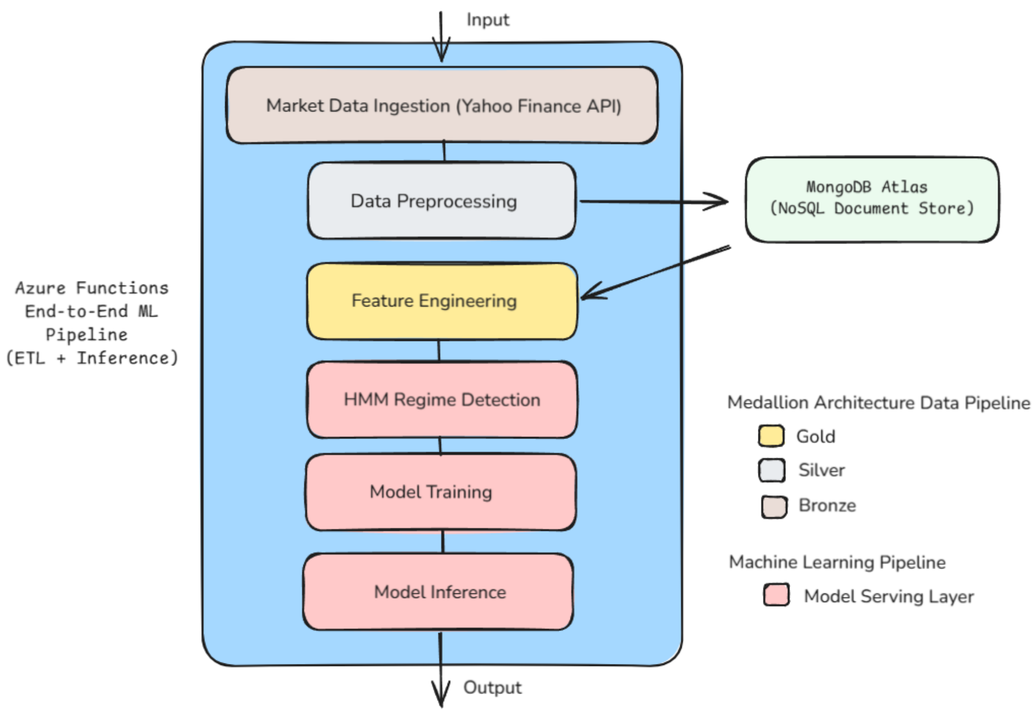

Fig. 24: End-to-end system architecture of the proposed regime-aware predictive framework.

A Additional Figures

A.1 Hyperparameter Search Results

λ	α	Depth	Colsample	$\text{Precision}_{\text{weighted}}$	$\text{F1}_{\text{weighted}}$
0.1	0.00	4	0.8	0.474	0.461
0.1	0.00	4	0.9	0.491	0.483
0.1	0.00	4	1.0	0.466	0.455
0.1	0.05	4	0.8	0.491	0.483
0.1	0.05	4	0.9	0.440	0.433
0.1	0.05	4	1.0	**0.497**	0.482
0.2	0.00	4	0.8	0.485	0.472
0.2	0.00	4	0.9	0.495	0.486
0.2	0.00	4	1.0	0.493	0.482
0.2	0.05	4	0.8	0.482	0.475
0.2	0.05	4	0.9	0.493	**0.491**
0.2	0.05	4	1.0	0.479	0.475
0.5	0.00	4	0.8	0.471	0.463
0.5	0.00	4	0.9	0.459	0.459
0.5	0.00	4	1.0	0.484	0.476
0.5	0.05	4	0.8	0.446	0.441
0.5	0.05	4	0.9	0.484	0.480
0.5	0.05	4	1.0	0.475	0.458
0.1	0.00	6	0.8	0.436	0.430
0.1	0.00	6	0.9	0.439	0.432
0.1	0.00	6	1.0	0.445	0.435
0.1	0.05	6	0.8	0.444	0.440
0.1	0.05	6	0.9	0.482	0.477
0.1	0.05	6	1.0	0.438	0.431
0.2	0.00	6	0.8	0.441	0.437
0.2	0.00	6	0.9	0.482	0.472
0.2	0.00	6	1.0	0.482	0.478
0.2	0.05	6	0.8	0.438	0.432
0.2	0.05	6	0.9	0.462	0.459
0.2	0.05	6	1.0	0.455	0.451
0.5	0.00	6	0.8	0.464	0.460
0.5	0.00	6	0.9	0.471	0.467
0.5	0.00	6	1.0	0.458	0.452
0.5	0.05	6	0.8	0.453	0.453
0.5	0.05	6	0.9	0.455	0.449
0.5	0.05	6	1.0	0.466	0.458
0.1	0.00	8	0.8	0.421	0.419
0.1	0.00	8	0.9	0.449	0.449
0.1	0.00	8	1.0	0.438	0.434
0.1	0.05	8	0.8	0.438	0.435
0.1	0.05	8	0.9	0.430	0.424
0.1	0.05	8	1.0	0.437	0.432
0.2	0.00	8	0.8	0.430	0.427
0.2	0.00	8	0.9	0.450	0.434
0.2	0.00	8	1.0	0.439	0.433
0.2	0.05	8	0.8	0.447	0.446
0.2	0.05	8	0.9	0.453	0.446
0.2	0.05	8	1.0	0.417	0.410
0.5	0.00	8	0.8	0.470	0.465
0.5	0.00	8	0.9	0.435	0.429
0.5	0.00	8	1.0	0.431	0.430
0.5	0.05	8	0.8	0.452	0.450
0.5	0.05	8	0.9	0.476	0.464
0.5	0.05	8	1.0	0.456	0.440

TABLE 11: LightGBM hyperparameter performance under rolling-window grid search.

Learning Rate	Depth	$\text{Precision}_{\text{weighted}}$	$\text{F1}_{\text{weighted}}$
0.03	4	**0.522**	**0.453**
0.01	4	0.486	0.444
0.02	4	0.482	0.430
0.05	6	0.456	0.430
0.01	6	0.480	0.435
0.03	6	0.443	0.417
0.02	6	0.445	0.417
0.05	7	0.402	0.382
0.01	7	0.447	0.414
0.03	7	0.399	0.377
0.02	7	0.439	0.412
0.05	8	0.394	0.385
0.03	8	0.421	0.404
0.01	8	0.424	0.403
0.02	8	0.370	0.355

TABLE 12: CatBoost hyperparameter performance under rolling-window grid search.

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